Delhivery: Feature Engineering

Feature engineering for delivery-time modeling and operational reliability

- Case Score: 77.0/100
- Status: Completed
- Due Date: 29 Nov 2025
- Report format aligned to WOOLF-style thesis expectations.

Business Context

- Delhivery handles high-volume logistics where route-level planning error directly impacts SLA compliance.
- The case focuses on improving predictive readiness through engineered operational features.
- Objective: reduce time-estimation error and improve explainability for planning teams.

Objectives and Scope

- Create robust engineered features from time, route, and distance attributes.
- Compare baseline and engineered models using MAE, RMSE, and R2.
- Translate feature-level signals into operational actions.

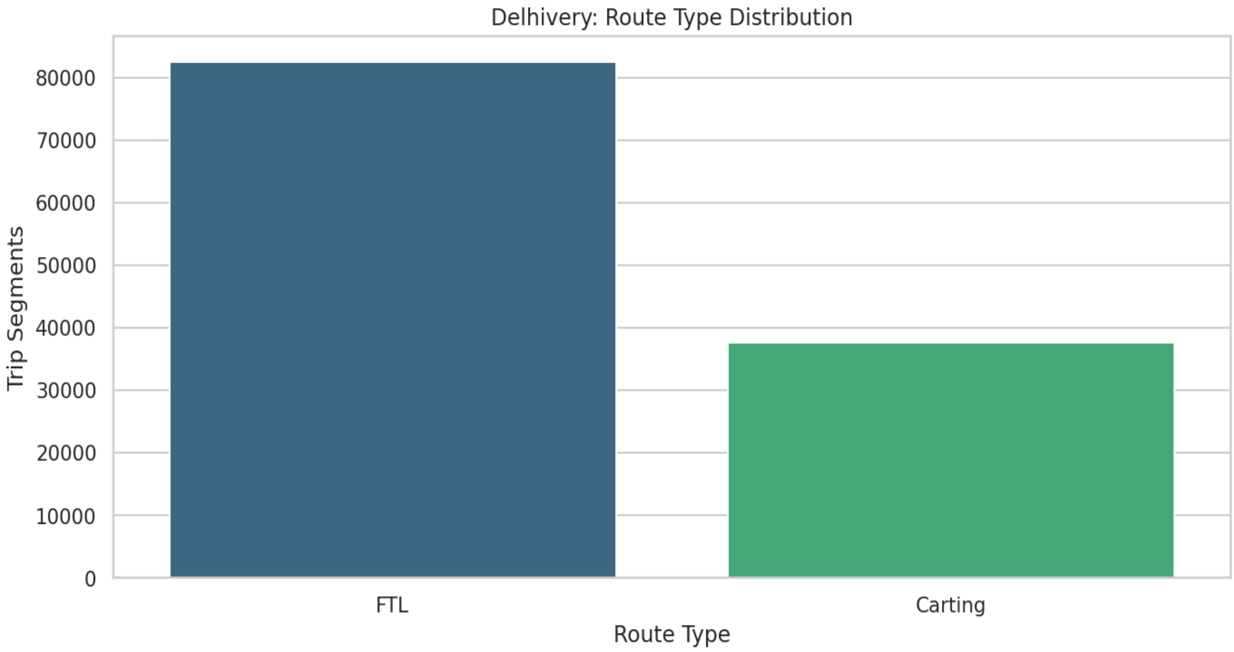
Dataset Overview

Attribute	Value
Rows	144,867
Columns	30
Primary Target	segment_actual_time
Core Route Variable	route_type
Missing Value %	0.01%

Data Quality and Preparation

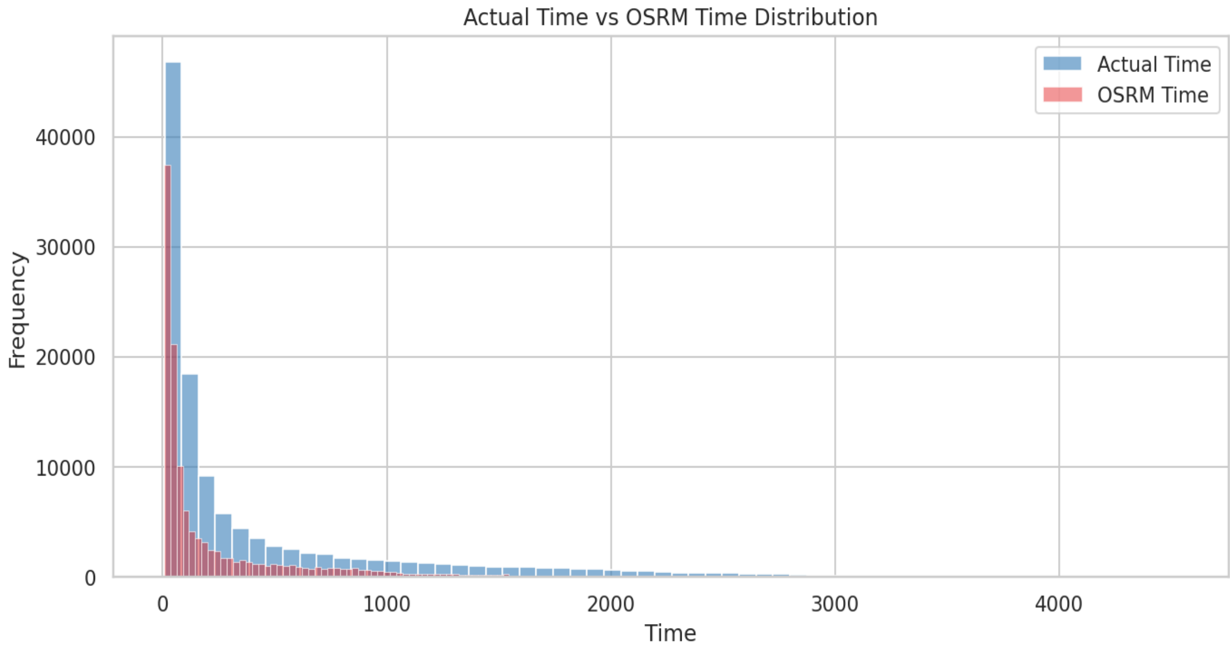
- Datetime fields were parsed with coercion to isolate invalid records.
- Numeric fields were standardized to numeric dtype with robust missing handling.
- Outliers were retained but evaluated through residual and distribution diagnostics.

Route Type Distribution



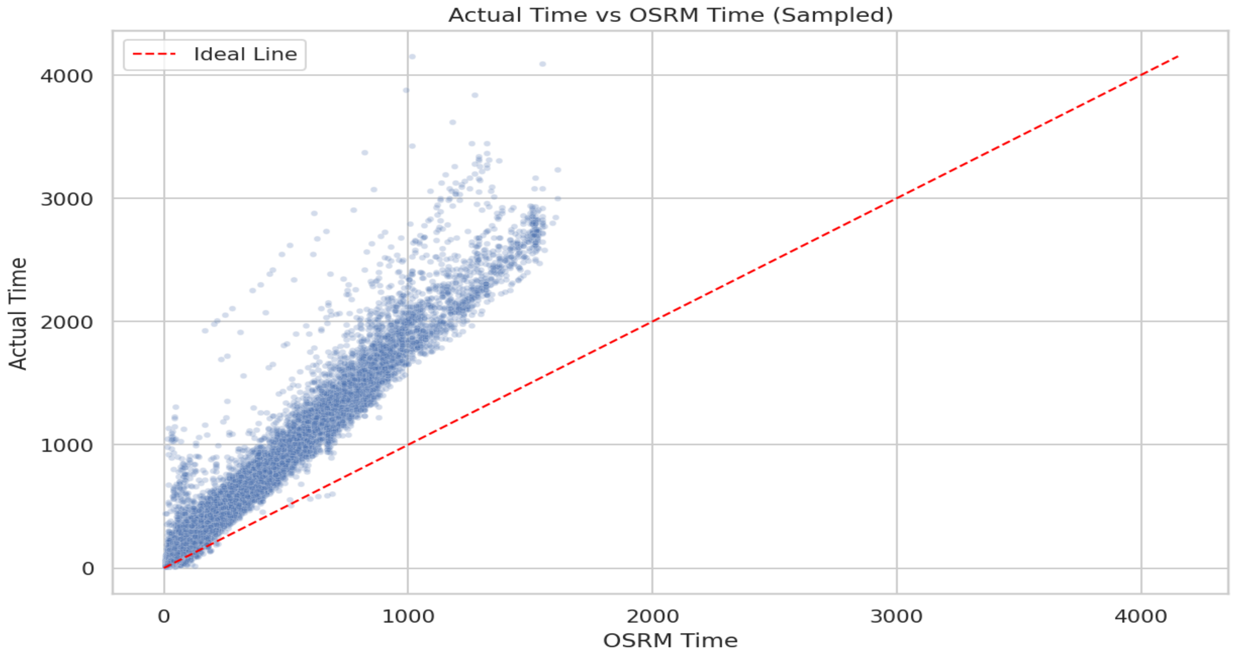
- This chart highlights the operational mix across route categories and helps prioritize optimization effort.

Actual vs OSRM Time Distribution



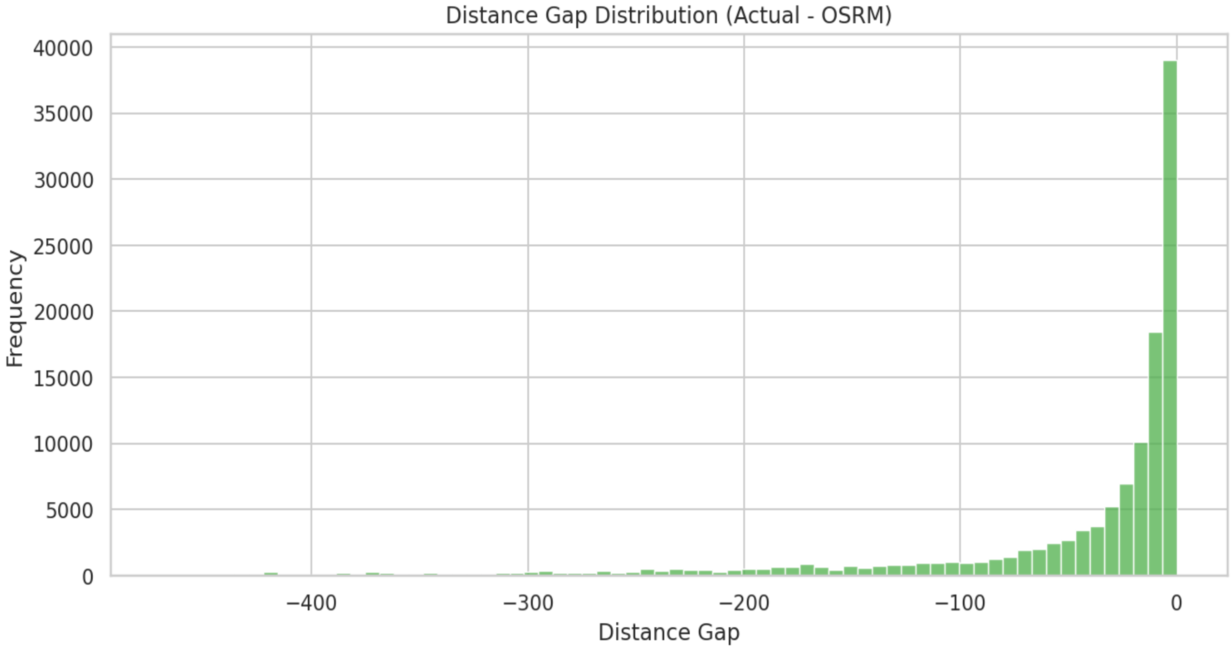
• The distribution gap indicates where planning estimates diverge from observed travel times.

Actual Time vs OSRM Time Scatter



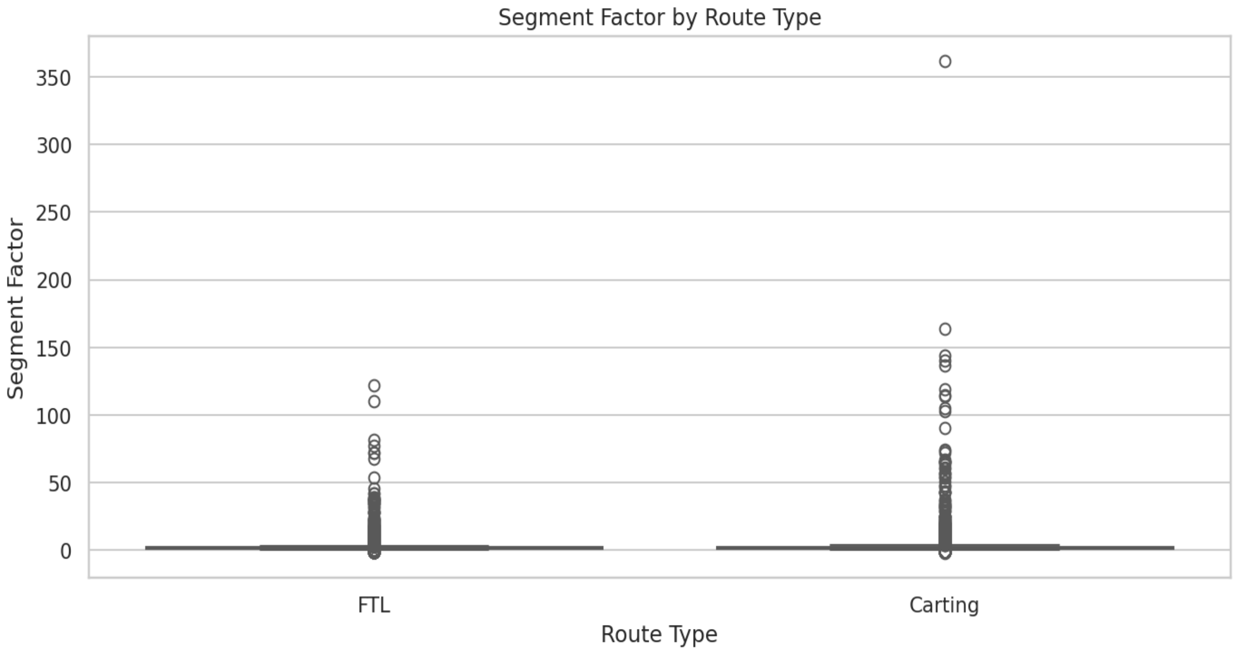
• Points above the diagonal mark segments where delivery took longer than route-engine estimates.

Distance Gap Distribution



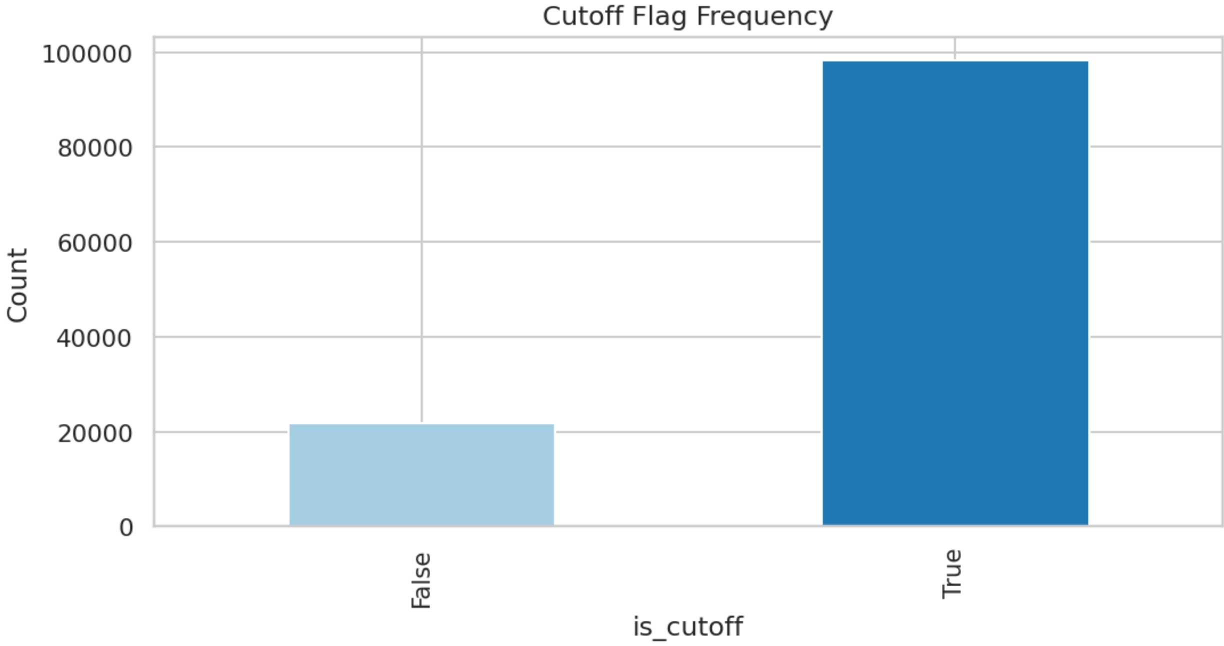
- Distance gap tracks route complexity and helps identify systematic mapping mismatches.

Segment Factor by Route Type



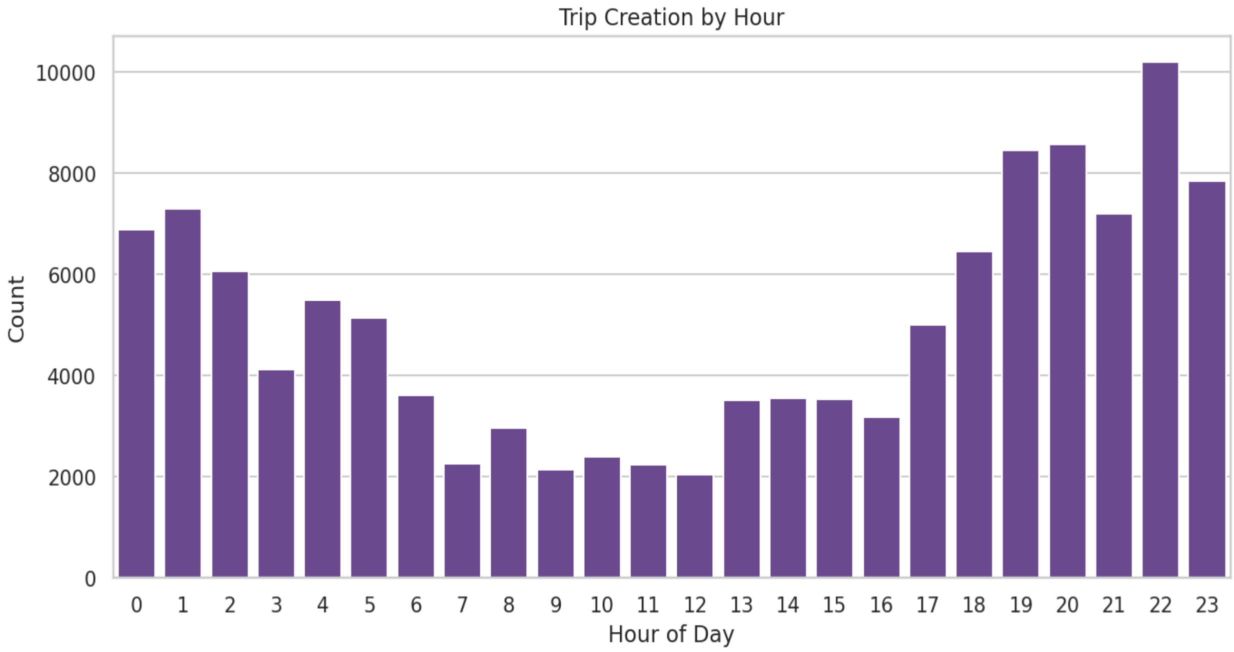
• Segment factor variability reveals route classes with unstable delivery performance.

Cutoff Flag Frequency



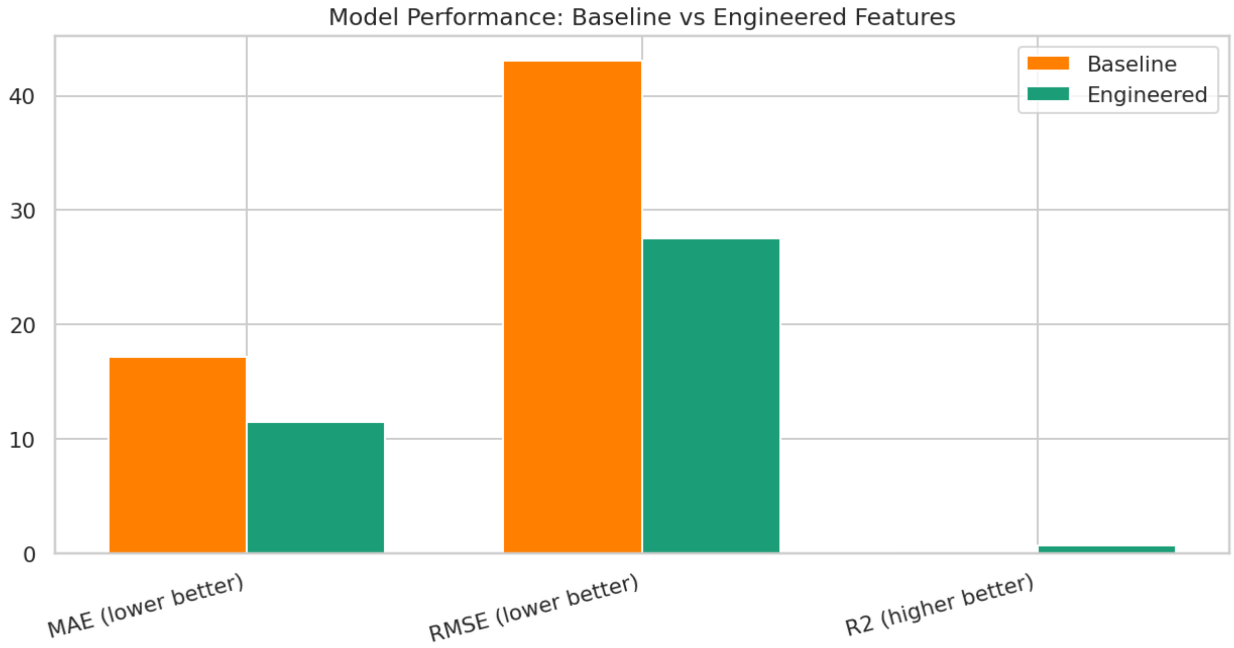
• Cutoff behavior supports SLA diagnostics and escalation planning.

Trip Creation by Hour



- Hourly concentration helps align resource allocation with demand peaks.

Model Performance Comparison



• Feature engineering materially improves predictive quality and operational forecast reliability.

Statistical and Reporting Summary

Metric	Result
Baseline MAE	17.18
Baseline RMSE	43.09
Baseline R2	0.1360
Engineered MAE	11.50
Engineered RMSE	27.48
Engineered R2	0.6485
R2 Improvement	0.5124

Key Insights

- Route-level timing gap and distance gap are strong explanatory factors.
- Engineered model offers lower error and improved variance explanation.
- Peak-hour concentration indicates capacity planning opportunities.

Business Recommendations

- Productionize engineered features in the scoring pipeline for ETA prediction.
- Use route-type-specific calibration factors for dispatch planning.
- Create monitoring alerts for rising residual error by route segment.

Implementation Roadmap

- 0-30 days: lock feature schema and baseline monitoring.
- 30-60 days: integrate engineered model with route planning dashboard.
- 60-90 days: segment-level SLA policy and auto-retraining trigger rollout.

Risks and Limitations

- Data drift in route behavior can reduce model stability.
- Missing or delayed scan updates may weaken feature reliability.
- Operational policy changes can shift target distribution.

Reproducibility and References

- Delhivery case dataset from Scaler cloudfront source
- Notebook lineage: Delhivery_Case_Study.ipynb
- Method stack: pandas, seaborn, scipy, scikit-learn

Chapter Closure

- Feature engineering substantially improves model readiness for route-time prediction.
- Operational decision support improves when feature importance is explicitly tracked.

Yulu: Hypothesis Testing

Statistical inference for rental demand behavior

- Case Score: 90.0/100
- Status: Completed
- Due Date: 11 Nov 2025
- Report format aligned to WOOLF-style thesis expectations.

Business Context

• Yulu observed revenue pressure and required quantified demand-driver understanding.

• Case objective is to test whether season, weather, and working-day behavior significantly alter rental count.

• Analysis outcomes are intended for operations planning and dynamic resource allocation.

Objectives and Scope

- Test working-day effect using two-sample t-test.
- Test season and weather effects using one-way ANOVA.
- Test dependency between season and weather using Chi-square.

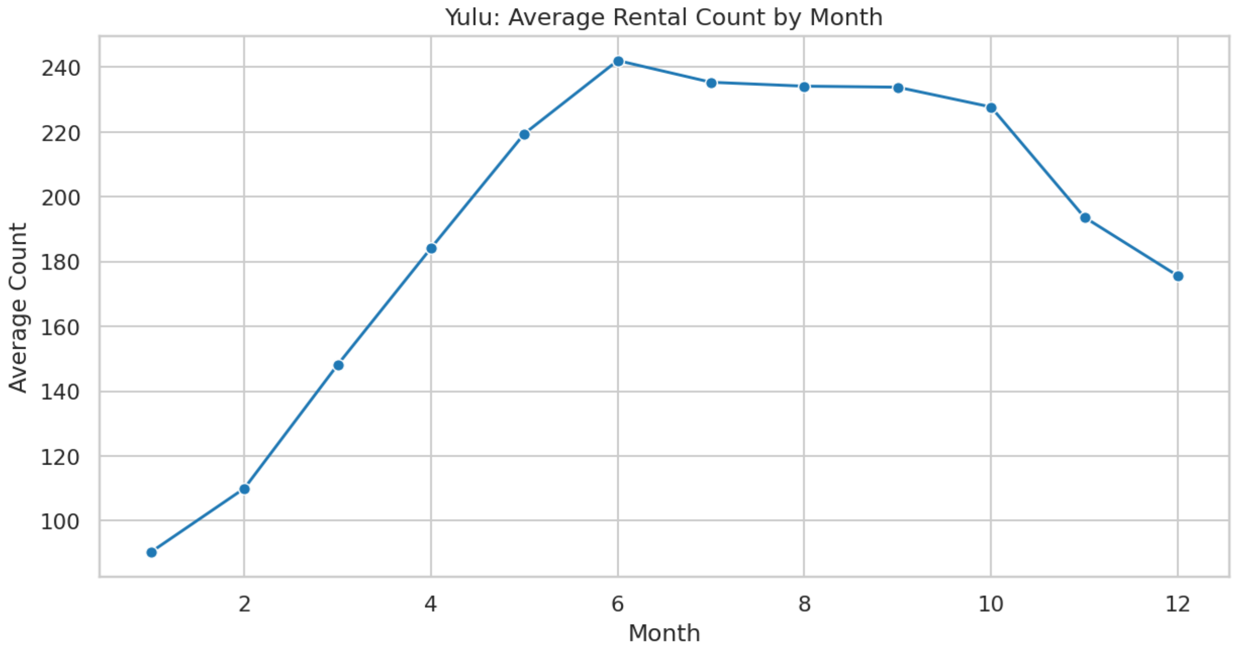
Dataset Overview

Attribute	Value
Rows	10,886
Columns	15
Target	count
Datetime Coverage	2011-01-01 to 2012-12-19
Missing Value %	0.00%

Data Quality and Preparation

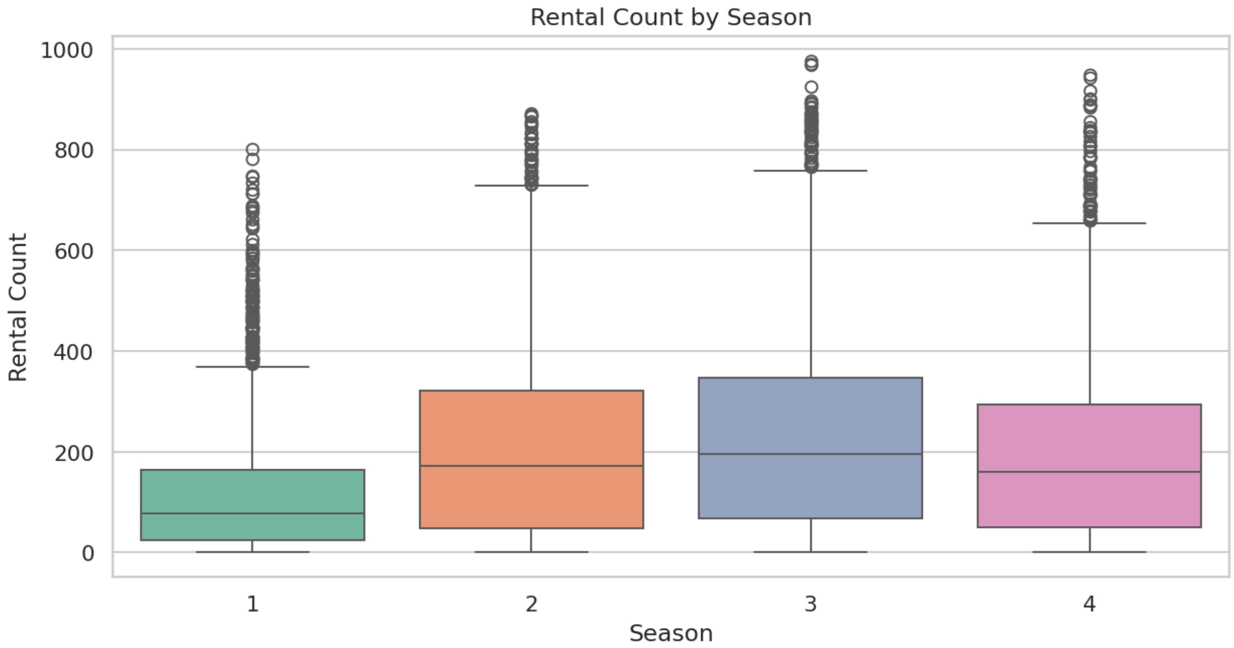
- Datetime field parsed and enriched into hour and month dimensions.
- No major missingness affecting inference columns.
- Statistical tests executed with explicit null/alternative framing.

Average Rental Count by Month



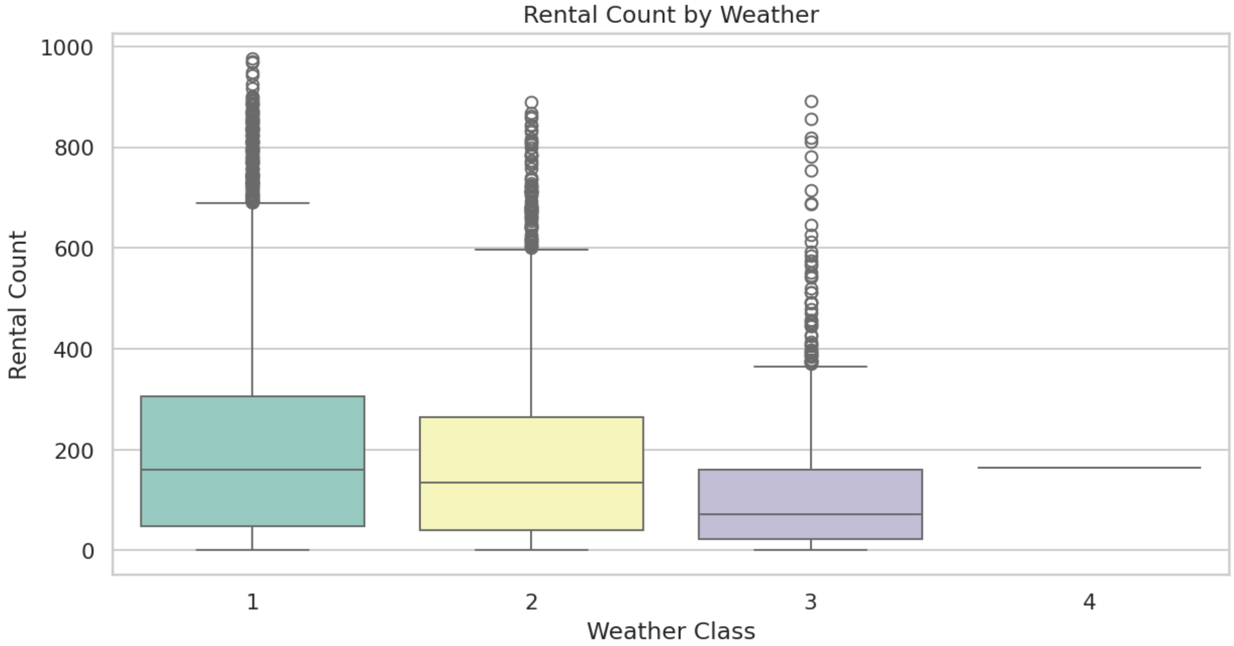
• Monthly trend indicates seasonality and supports demand planning.

Rental Distribution by Season



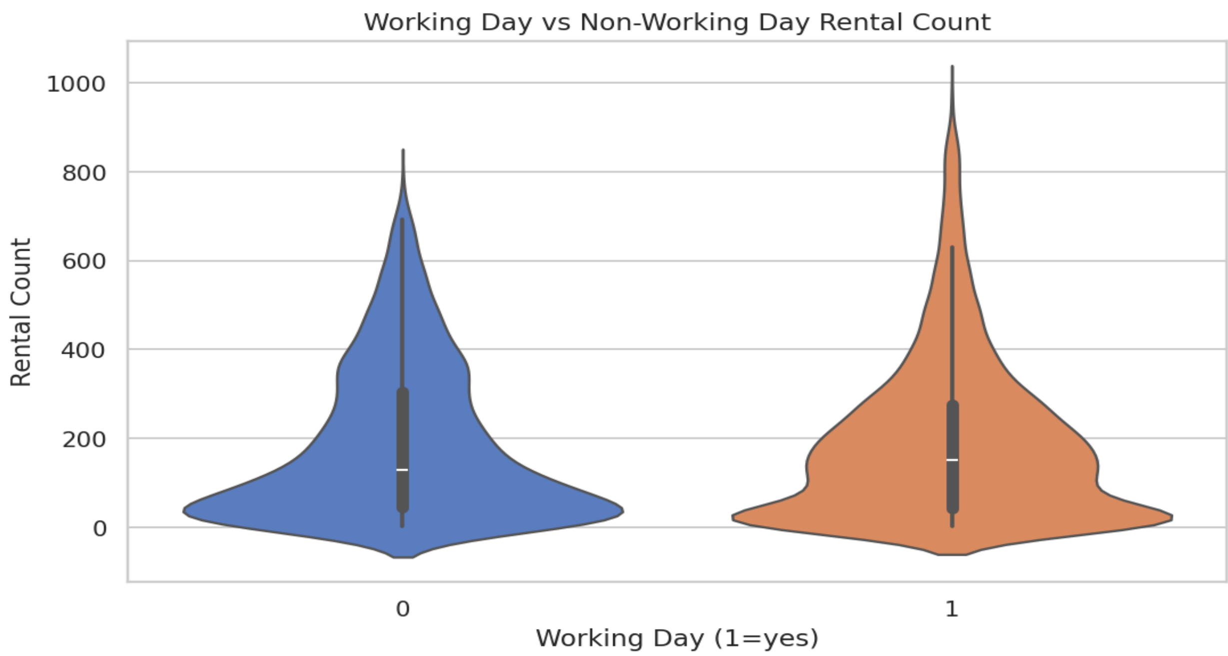
• Seasonal spread validates demand variation across climate cycles.

Rental Distribution by Weather



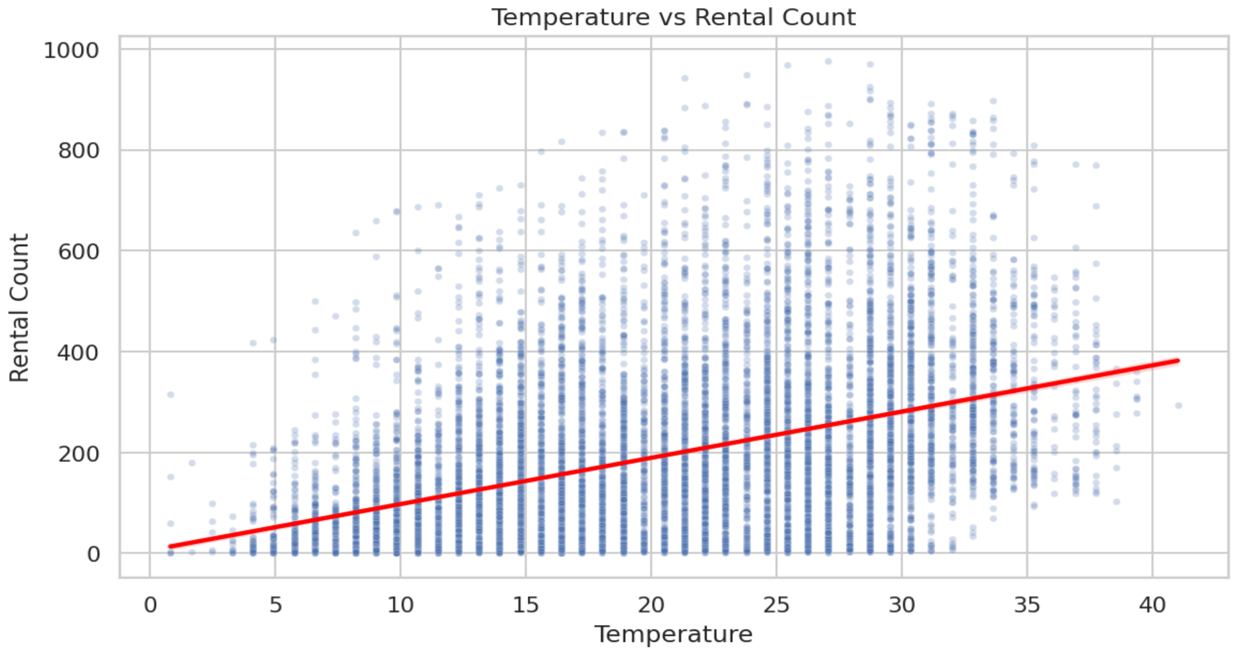
• Demand weakens in severe weather, reinforcing weather-aware operations.

Working Day Effect on Rentals



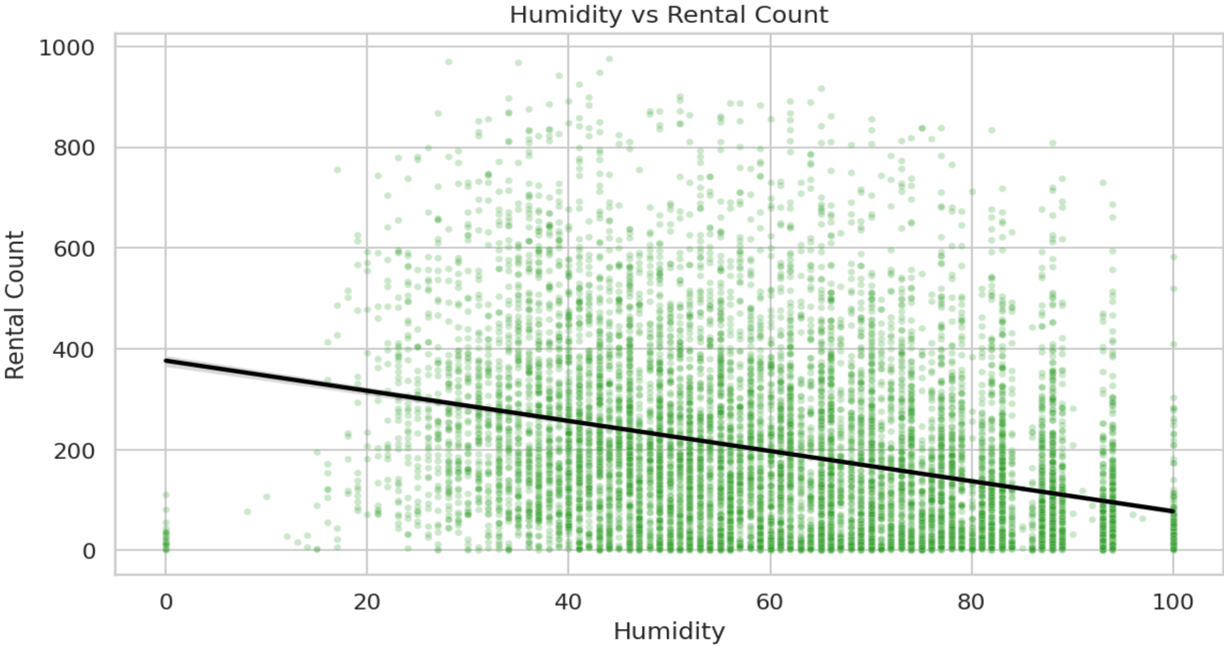
• Distribution overlap supports hypothesis-test interpretation for working day impact.

Temperature and Rental Count Relationship



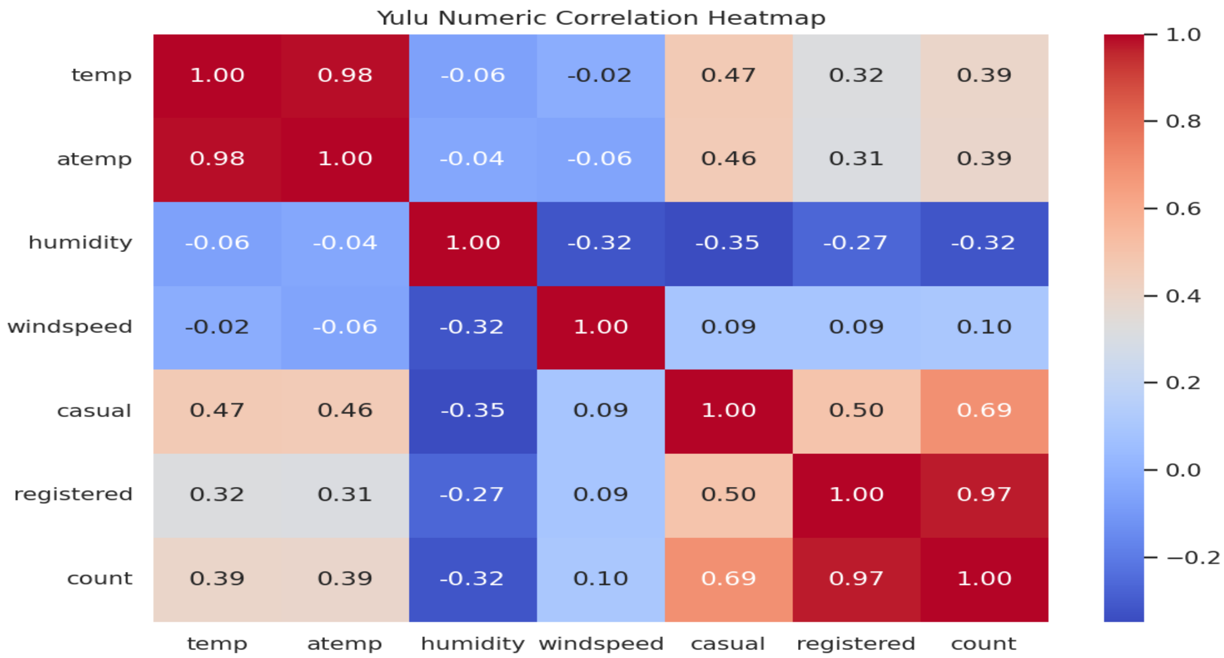
• Demand tends to increase in favorable temperature ranges.

Humidity and Rental Count Relationship



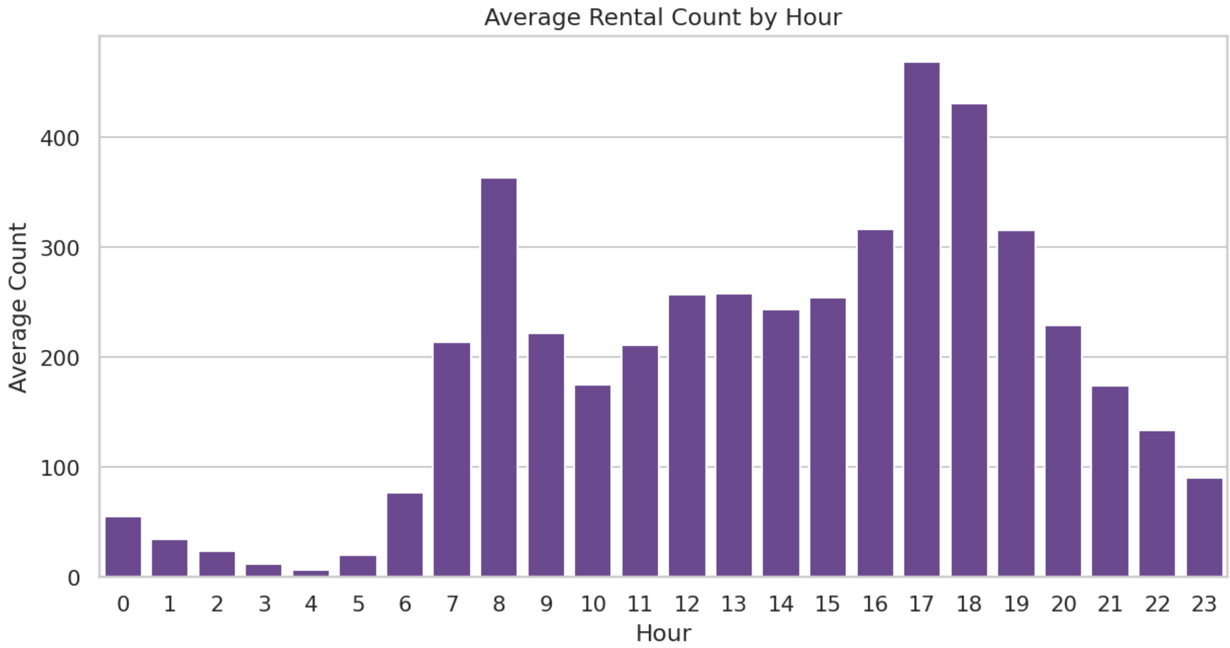
• Higher humidity bands are associated with lower rental intensity.

Correlation Heatmap



• Registered and casual rider behavior strongly contributes to total count.

Hourly Rental Profile



• Hourly demand profile supports shift-wise vehicle allocation strategy.

Statistical and Reporting Summary

Metric	Result
t-test p-value (working day)	0.216403
ANOVA p-value (season)	6.16484e-149
ANOVA p-value (weather)	5.48207e-42
Chi-square p-value (season-weather)	1.54993e-07
Mean count (working day=1)	193.01
Mean count (working day=0)	188.51

Key Insights

- Season and weather show statistically significant rental differences.
- Working-day effect is weaker relative to climate-related variables.
- Demand behavior confirms need for weather-aware fleet strategy.

Business Recommendations

- Adopt weather-sensitive dispatch and pricing rules.
- Prioritize high-demand seasonal windows for inventory and promotions.
- Integrate inferential monitoring in recurring planning reviews.

Implementation Roadmap

- 0-30 days: deploy weather/season dashboard and alerts.
- 30-60 days: run scenario-based staffing and fleet simulations.
- 60-90 days: align campaign and operations playbooks by demand cluster.

Risks and Limitations

- External shocks (events, policy, fuel costs) are not modeled directly.
- Hypothesis tests capture association, not full causal structure.
- Operational decisions need continuous back-testing with latest data.

Reproducibility and References

- Yulu dataset from Scaler cloudfront source
- Notebook lineage: Yulu.ipynb
- Method stack: scipy.stats, pandas, seaborn

Chapter Closure

- Hypothesis testing identified climate-sensitive demand structure with operational implications.
- Statistical evidence supports targeted planning over blanket policy approaches.

Walmart: Confidence Interval and CLT

Inferential analytics for purchase behavior segmentation

- Case Score: 94.0/100
- Status: Completed
- Due Date: 21 Oct 2025
- Report format aligned to WOOLF-style thesis expectations.

Business Context

- Walmart seeks statistically reliable spending insights across demographic segments.
- Case emphasis is uncertainty quantification and confidence-based decision-making.
- CLT demonstration informs interpretation robustness at scale.

Objectives and Scope

- Estimate and compare average transaction values across groups.
- Construct confidence intervals for actionable uncertainty bounds.
- Demonstrate sample-size impact using CLT simulation.

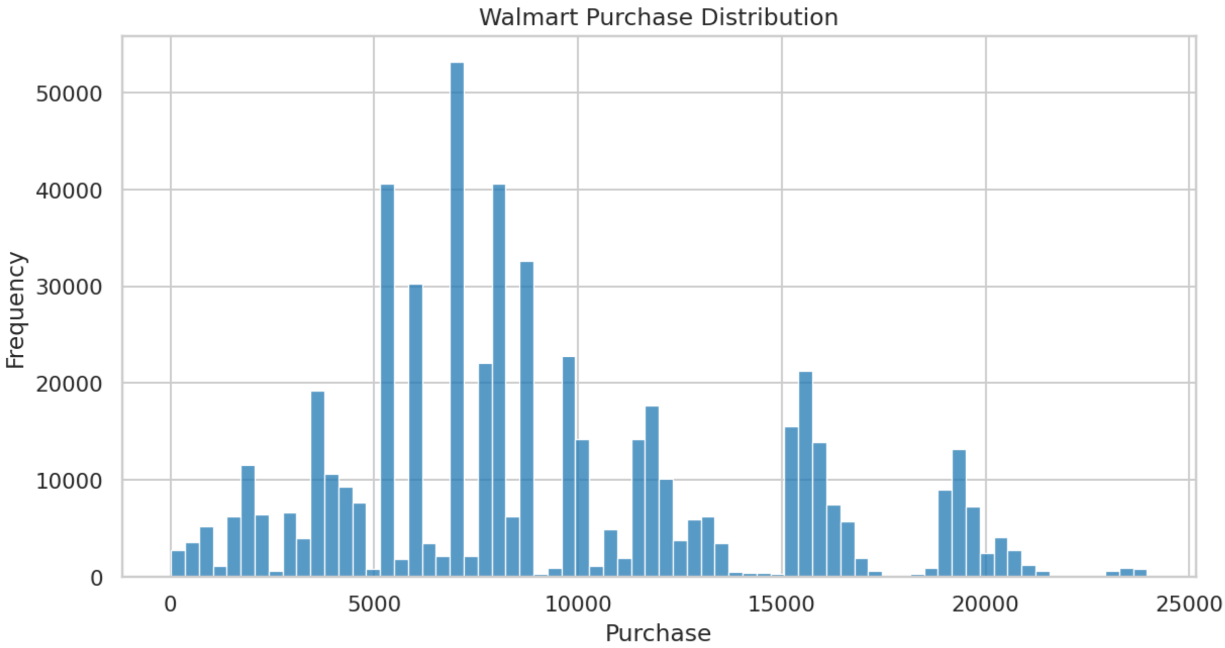
Dataset Overview

Attribute	Value
Rows	550,068
Columns	10
Target	Purchase
Unique Users	5,891
Missing Value %	0.00%

Data Quality and Preparation

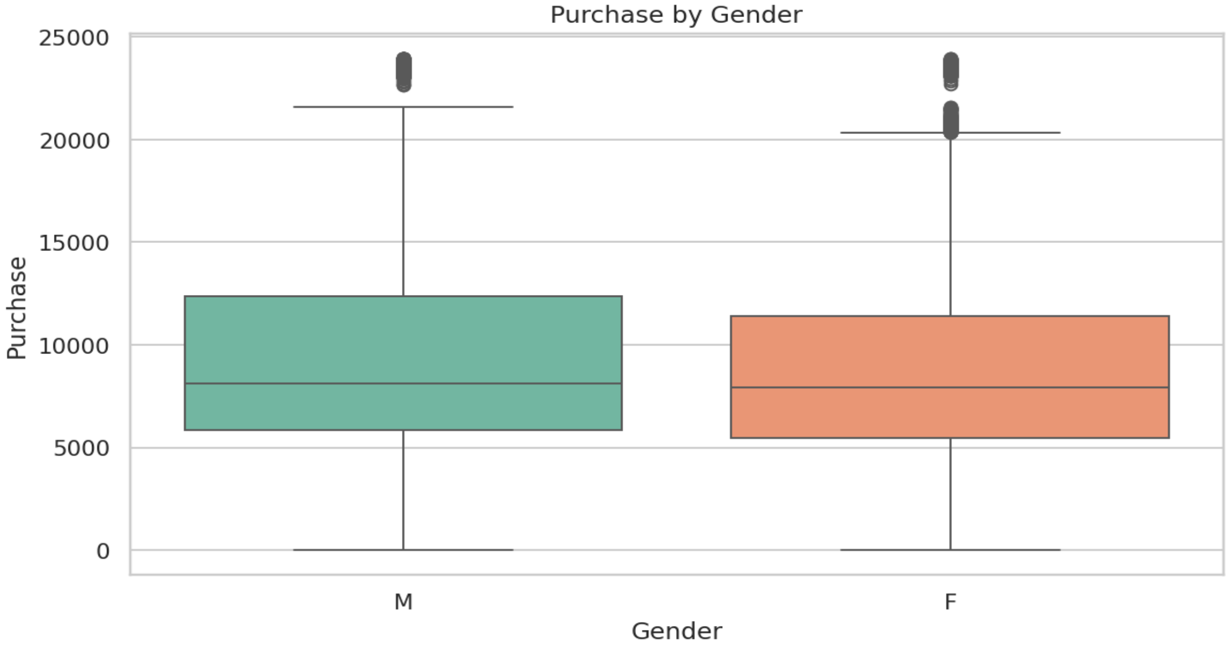
- Purchase field validated as numeric with invalid rows excluded.
- Category attributes standardized for grouped aggregation.
- Bootstrap and analytic CI methods used for robust inference.

Purchase Distribution



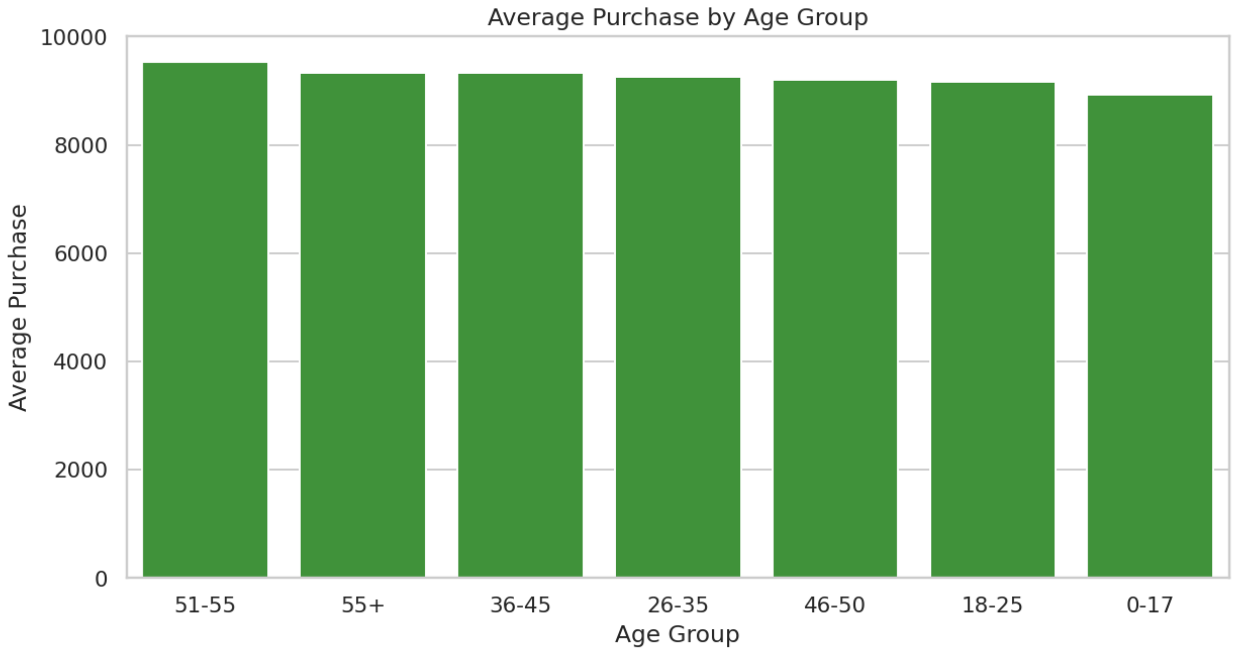
• Distribution is right-skewed with a long high-spend tail.

Purchase by Gender



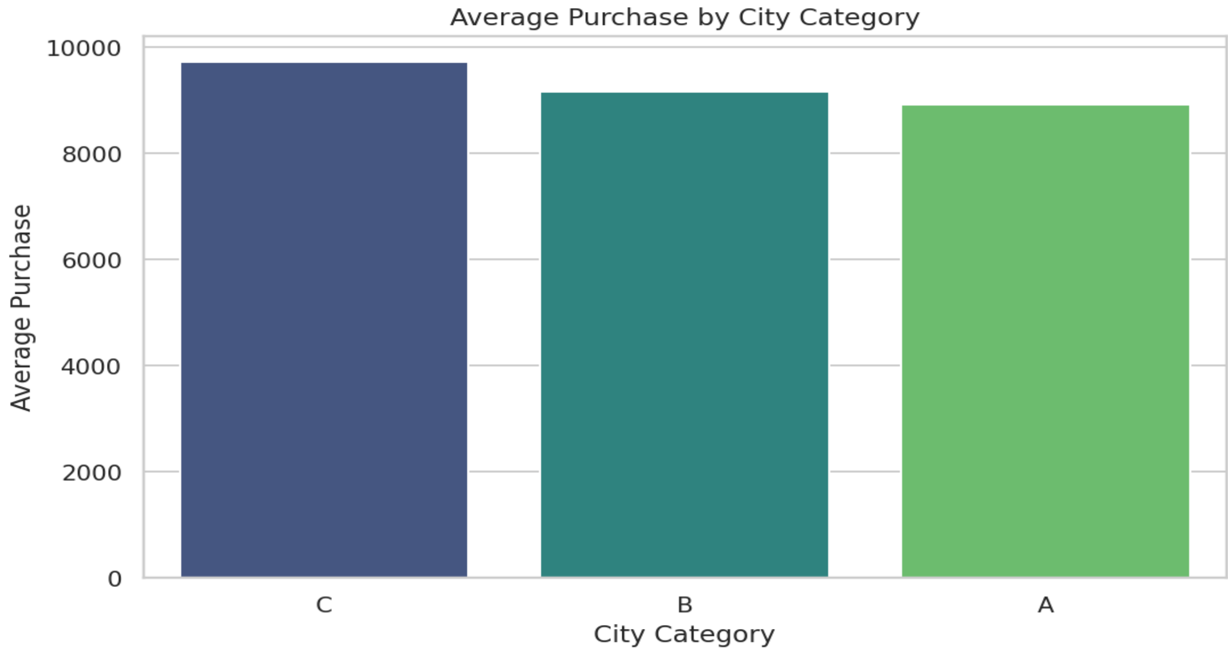
• Gender-level distribution supports differentiated campaign strategy.

Average Purchase by Age Group



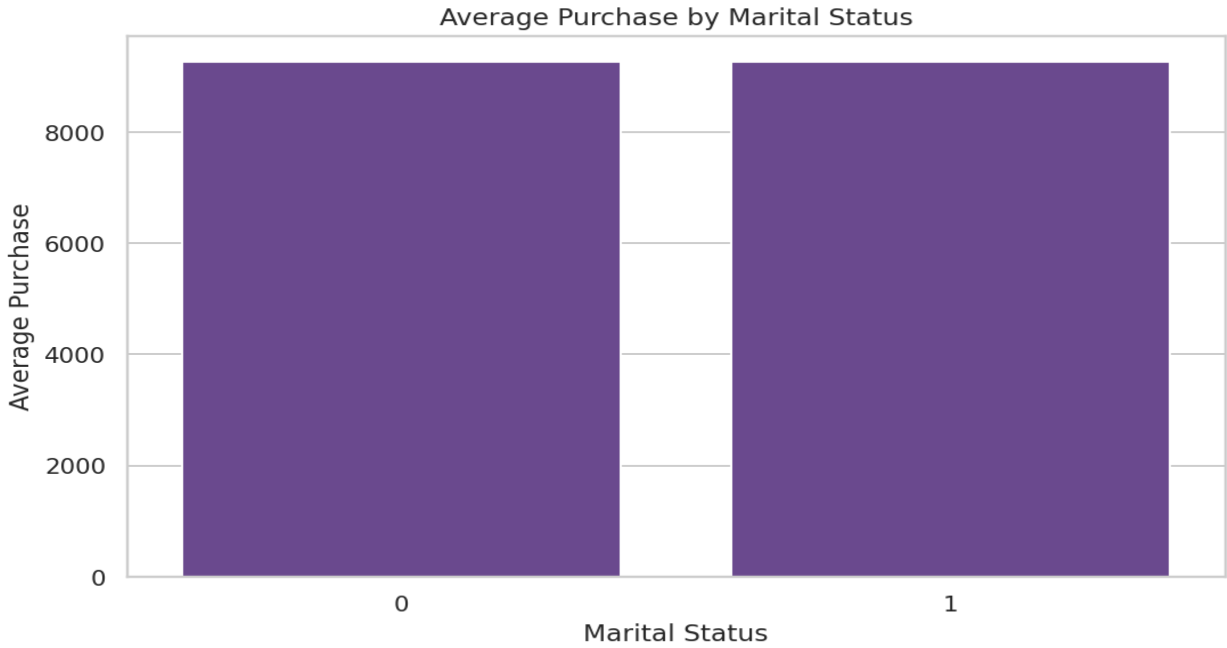
• Age bands reveal high-value cohorts for promotion prioritization.

Average Purchase by City Category



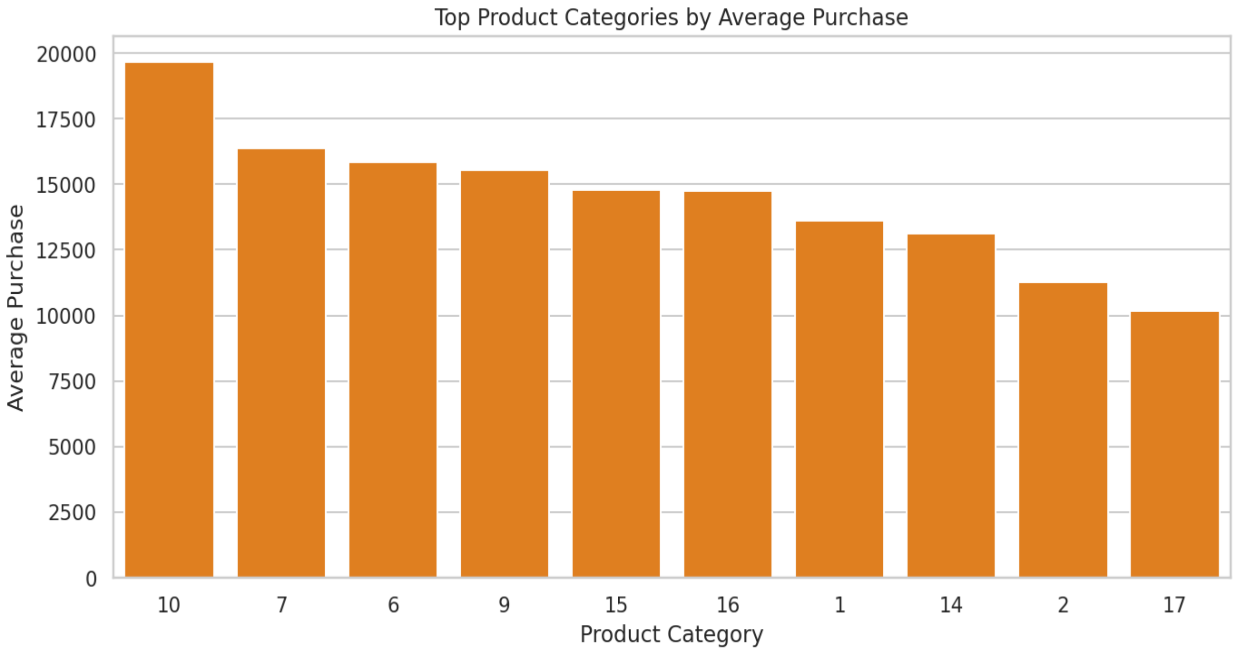
• City category segmentation informs inventory and channel planning.

Average Purchase by Marital Status



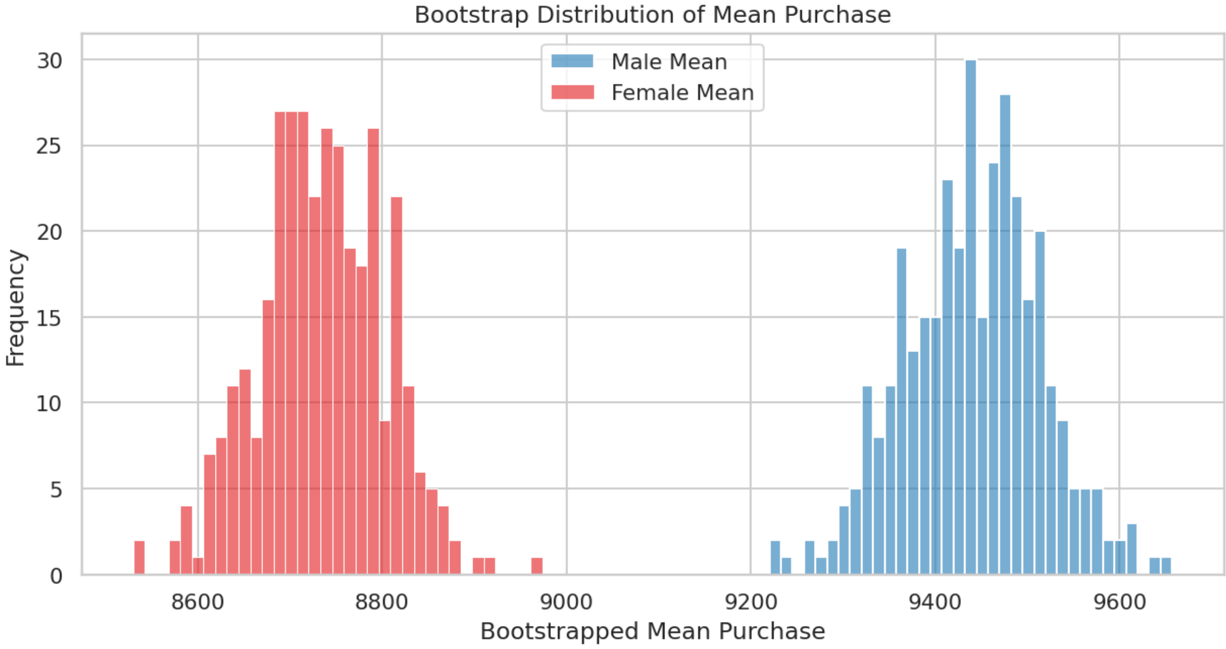
• Marital-status differences are smaller than gender or age segmentation effects.

Top Product Categories by Average Purchase



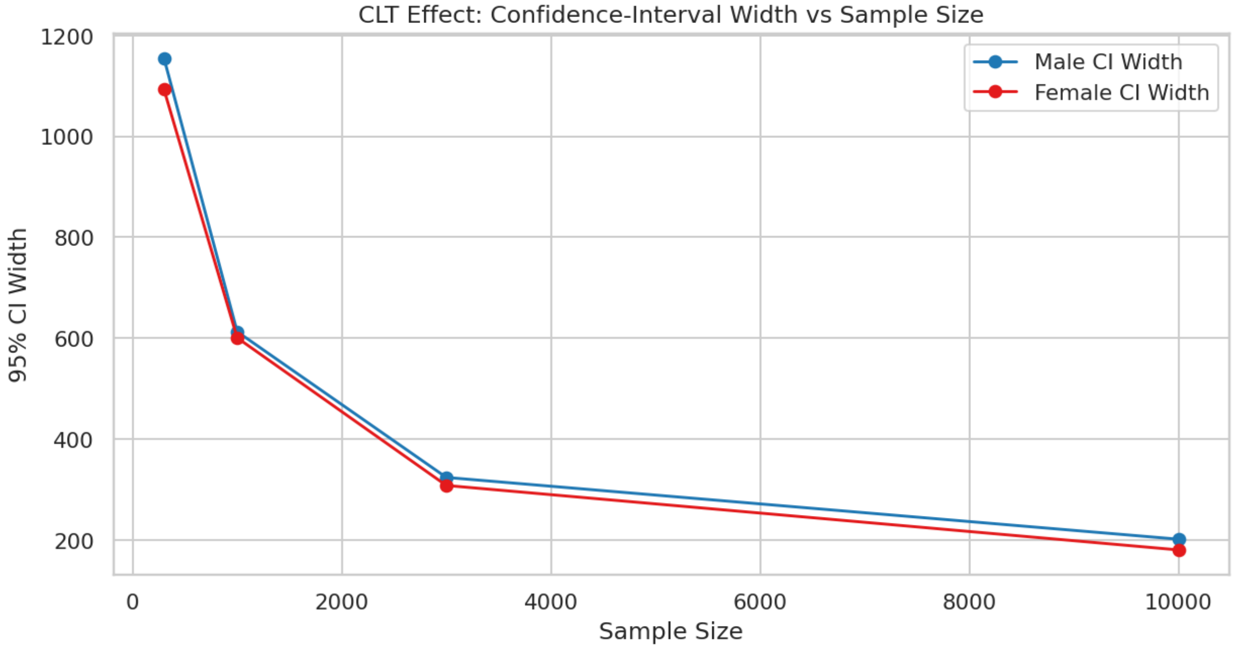
• High-value product categories can anchor premium offer design.

Bootstrap Mean Distribution by Gender



• Bootstrap distributions provide robust uncertainty estimation for group means.

CI Width Reduction with Sample Size (CLT)



• As sample size increases, uncertainty narrows, consistent with CLT behavior.

Statistical and Reporting Summary

Metric	Result
Mean Purchase (Male)	9,437.53
Mean Purchase (Female)	8,734.57
95% CI Male	[9,422.02, 9,453.03]
95% CI Female	[8,709.21, 8,759.92]
Overall Mean	9,263.97
Overall Median	8,047.00

Key Insights

- Gender and age segmentation reveal clearer spending contrasts than marital status.
- CI overlap analysis supports prioritization confidence in campaign targeting.
- Larger sample windows materially improve estimation precision.

Business Recommendations

- Deploy segment-specific promotional bundles for high-value cohorts.
- Use CI-based thresholds for marketing experiment acceptance.
- Prioritize age and city segmentation in demand planning dashboards.

Implementation Roadmap

- 0-30 days: integrate CI reporting into weekly commercial review.
- 30-60 days: launch segmented offers and monitor uplift by confidence bounds.
- 60-90 days: optimize allocation using segment-level predictive loops.

Risks and Limitations

- Non-demographic factors (seasonality, campaigns) may confound segment means.
- Static segmentation can decay as customer behavior shifts.
- Over-interpretation of significance without business effect size can mislead decisions.

Reproducibility and References

- Walmart dataset from case repository
- Notebook lineage: wallmart solution.pynb
- Method stack: scipy.stats, bootstrap simulation

Chapter Closure

- Inference layer improves decision confidence beyond point estimates.
- CLT framing provides operationally useful guidance on sample adequacy.

Aerofit: Descriptive Statistics and Probability

Customer profiling with descriptive and conditional probability analysis

- Case Score: 86.0/100
- Status: Completed
- Due Date: 28 Sep 2025
- Report format aligned to WOOLF-style thesis expectations.

Business Context

- AeroFit requires model-specific customer profiling across KP281, KP481, and KP781.
- The case emphasizes descriptive diagnostics and probability-based segment interpretation.
- Findings are used for positioning, upsell pathways, and campaign targeting.

Objectives and Scope

- Describe customer behavior by product segment.
- Estimate marginal and conditional purchase probabilities.
- Translate profile patterns into actionable GTM recommendations.

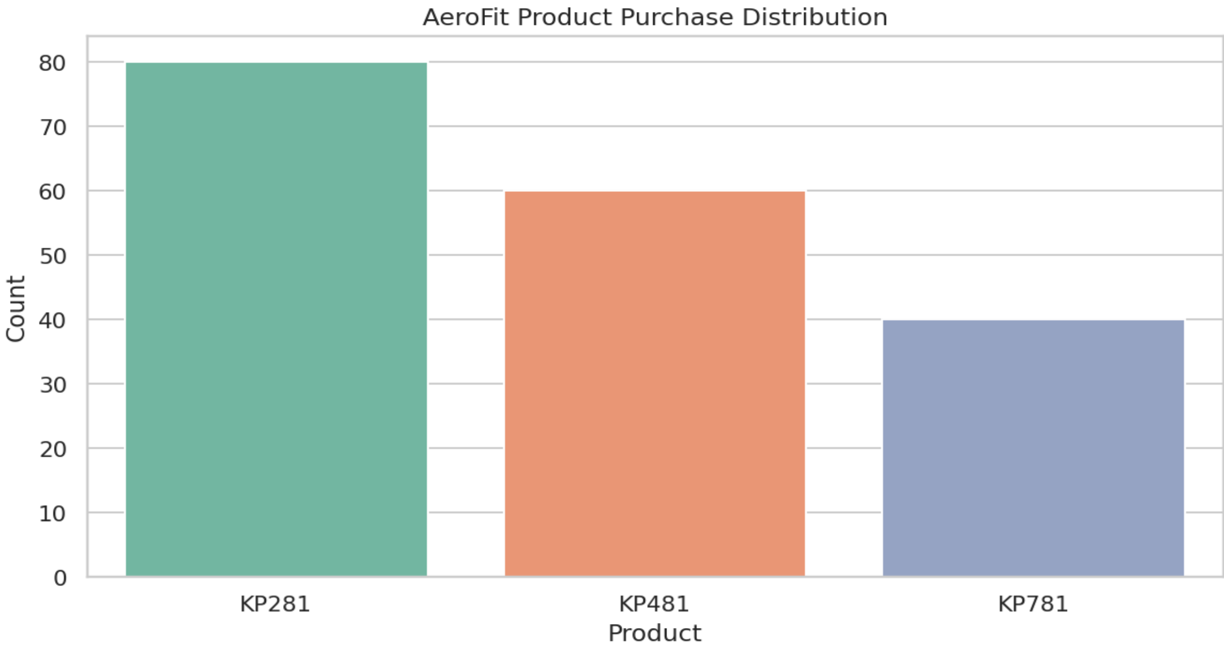
Dataset Overview

Attribute	Value
Rows	180
Columns	9
Products	KP281, KP481, KP781
Missing Value %	0.00%

Data Quality and Preparation

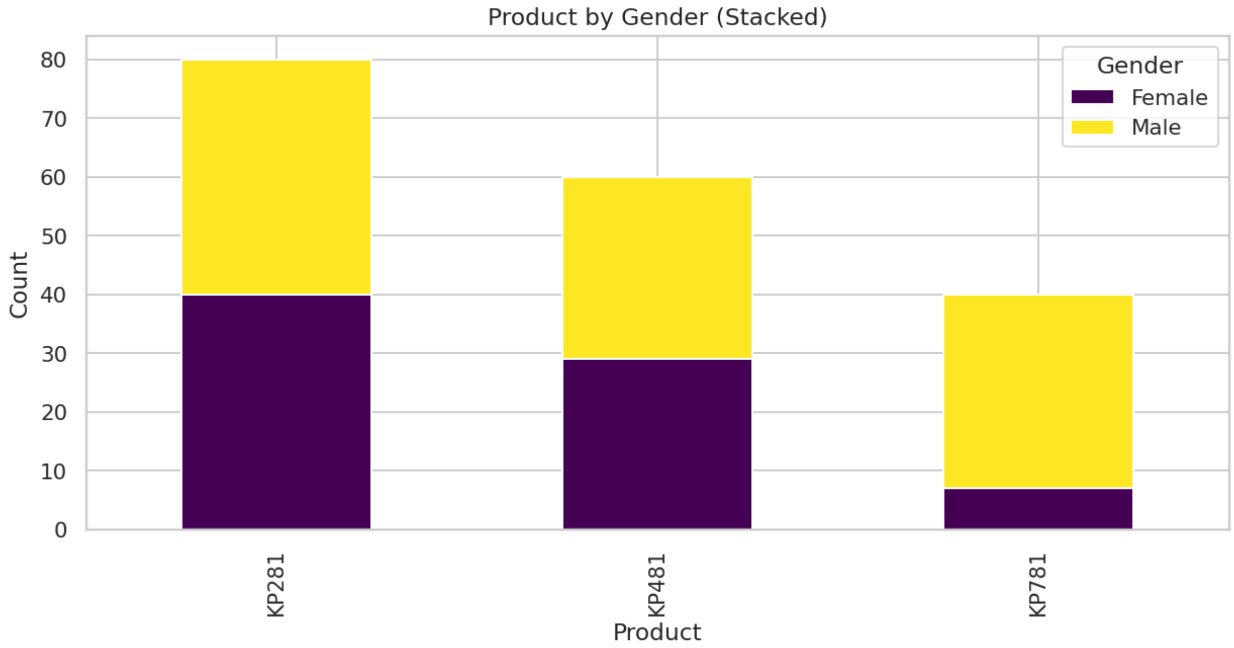
- Numeric fields standardized before summary and correlation analysis.
- Outlier patterns retained to preserve high-intensity customer behavior.
- Probability views built using normalized crosstab logic.

Product Purchase Distribution



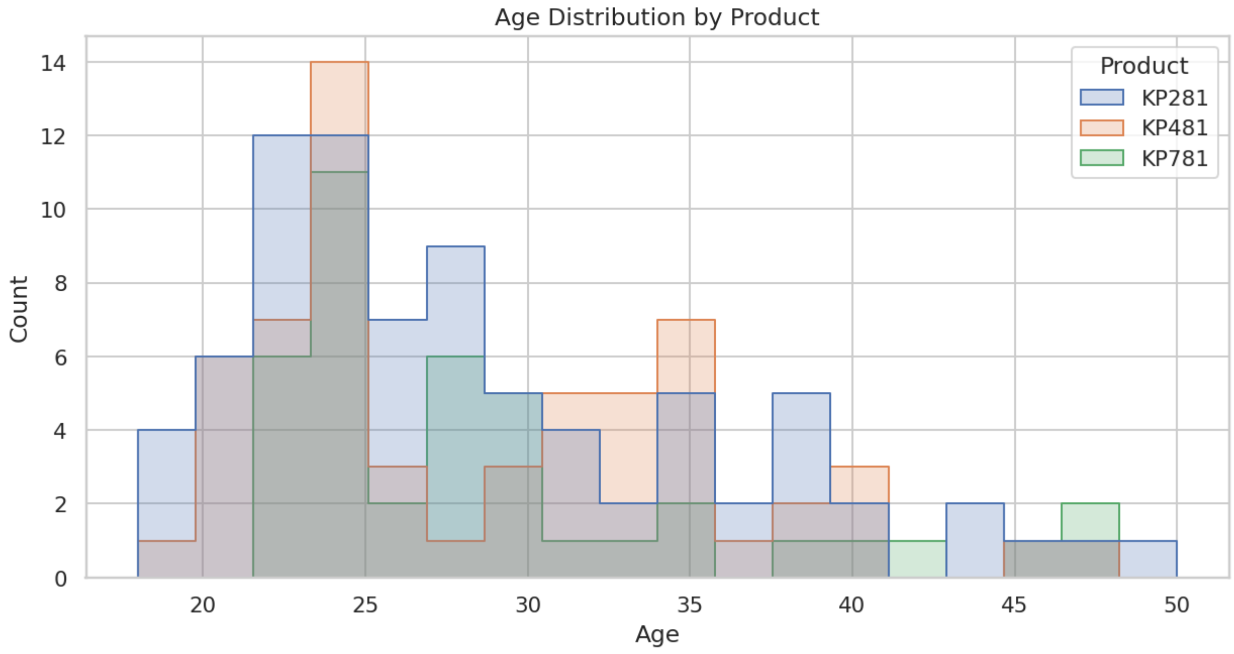
• Distribution across product tiers indicates entry, value, and premium demand mix.

Product by Gender



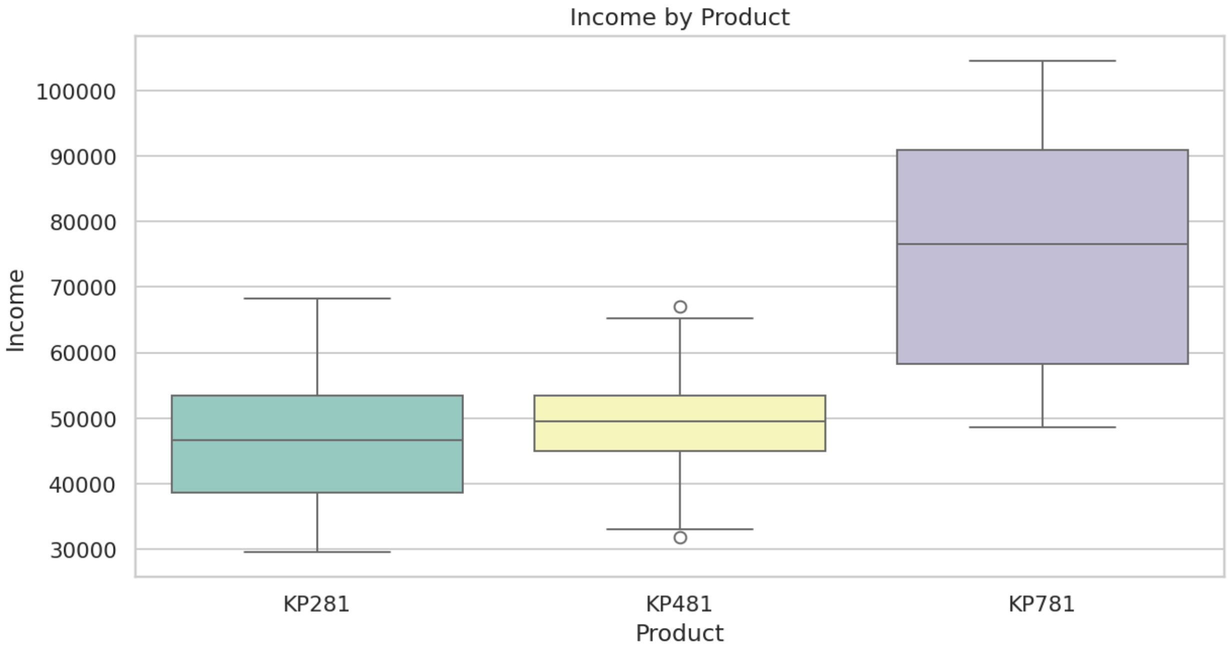
• Gender-product interactions help tailor communication and placement.

Age Distribution by Product



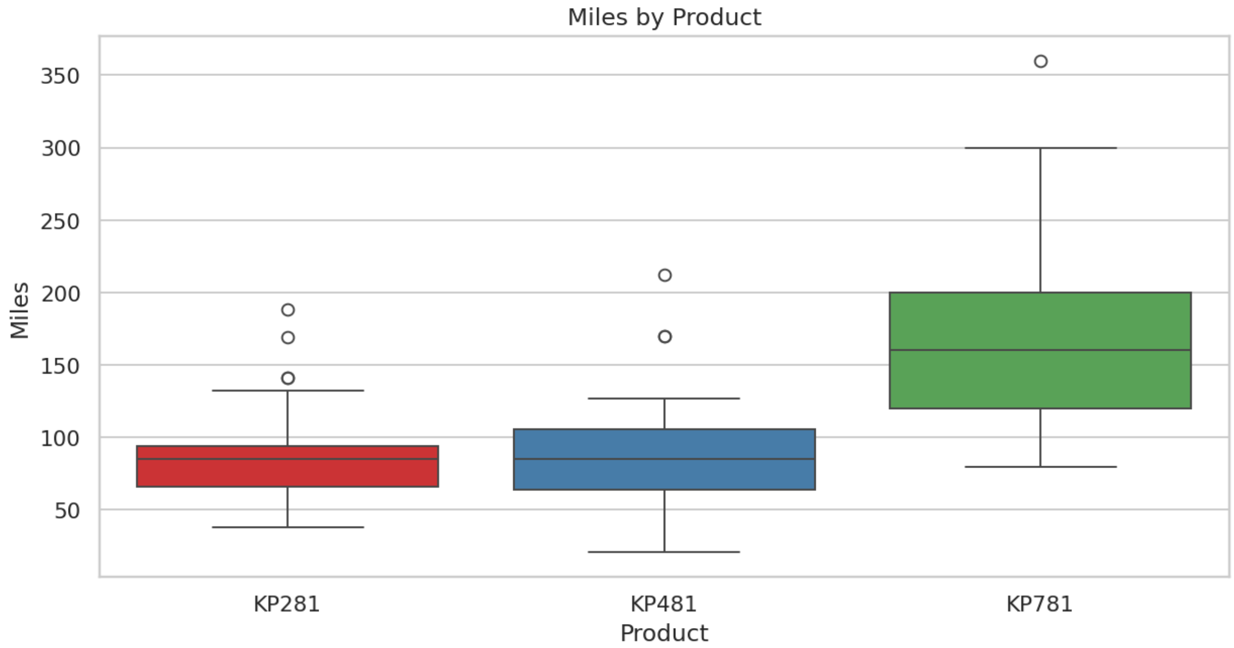
• Age concentration differs across product tiers and supports persona design.

Income Distribution by Product



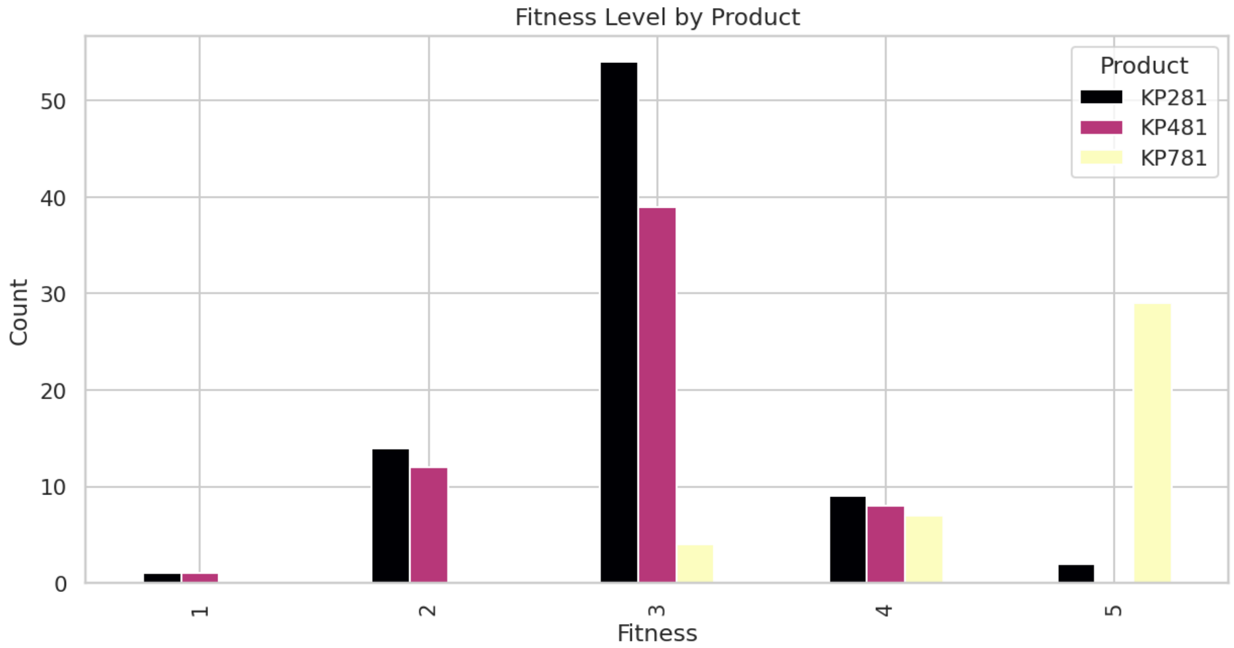
• Premium product preference aligns with higher income distribution bands.

Miles Distribution by Product



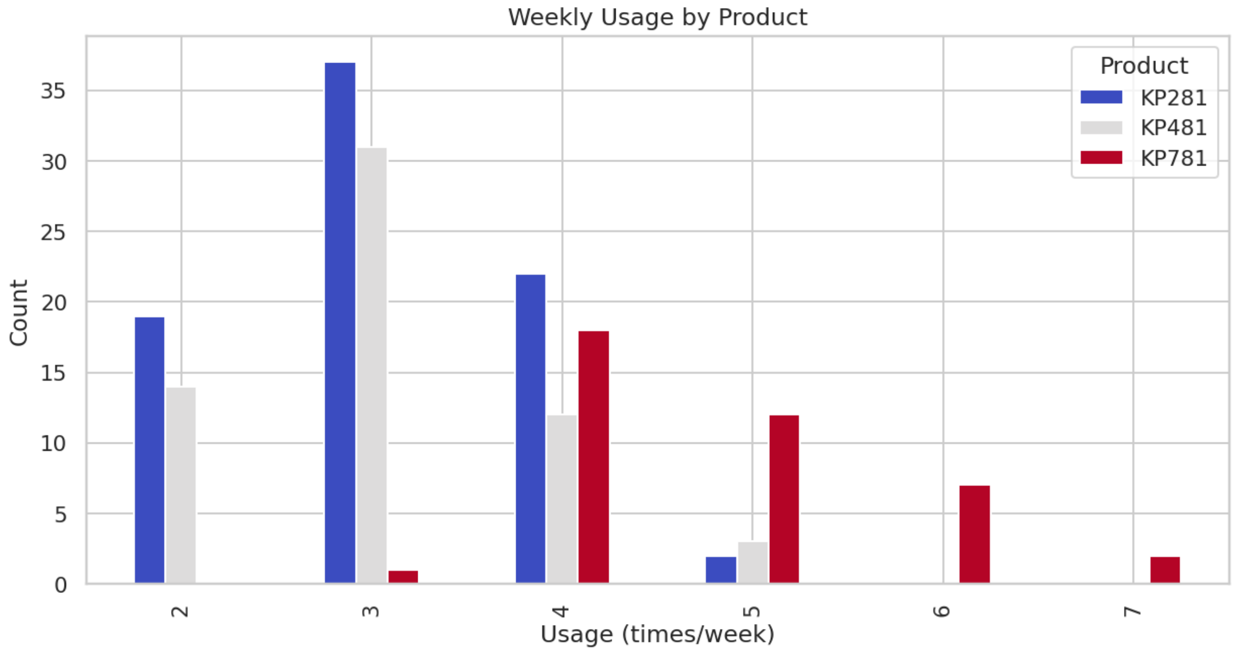
• Usage-intensity differences indicate performance-driven segment migration.

Fitness Level by Product



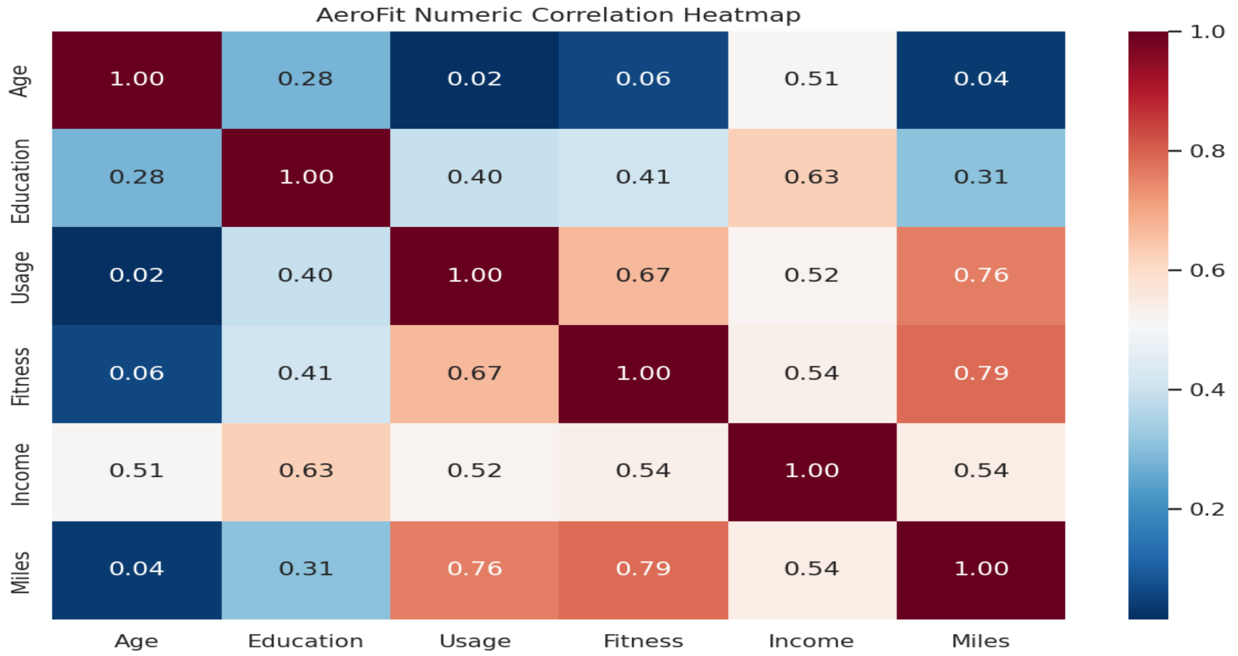
• Fitness level differentiates product affinity and informs upsell strategy.

Usage by Product



• Weekly-usage profiles provide actionable cues for product positioning.

Correlation Heatmap



• Correlated behavior dimensions support profile-based recommendation logic.

Statistical and Reporting Summary

Metric	Result
P(KP281)	0.444
P(KP481)	0.333
P(KP781)	0.222
P(KP781 | Male)	0.317
Average Income	53,719.58
Average Miles	103.19

Key Insights

- Customer profile differs by product tier on income, usage, and fitness dimensions.
- Conditional probabilities reveal strong segment affinity for specific products.
- Premium-tier buyers show distinct behavioral intensity and value potential.

Business Recommendations

- Run persona-specific messaging for each treadmill tier.
- Offer guided upgrade path from KP281 to KP481/KP781.
- Bundle service plans for high-usage customers to improve retention.

Implementation Roadmap

- 0-30 days: publish product-persona playbook and eligibility rules.
- 30-60 days: launch upgrade campaign pilots.
- 60-90 days: optimize conversion with probability-triggered nudges.

Risks and Limitations

- Small-sample behavior can overstate niche profile effects.
- Price and promotion context may shift preference patterns.
- Conditional probability is descriptive and should be revalidated periodically.

Reproducibility and References

- AeroFit dataset from repository CSV
- Notebook lineage: Aerofit case study solution.ipynb
- Method stack: descriptive stats, crosstab probability

Chapter Closure

- Probability-driven profiling enables sharper product targeting.
- Segment-centric recommendations improve conversion and upsell readiness.

Netflix: Data Exploration and Visualisation

EDA-driven content strategy diagnostics

- Case Score: 95.0/100
- Status: Completed
- Due Date: 26 Aug 2025
- Report format aligned to WOOLF-style thesis expectations.

Business Context

- Netflix requires data-backed content intelligence for catalog decisions.
- This case explores type, geography, release, rating, and genre patterns.
- Insights are intended for acquisition, production, and growth planning.

Objectives and Scope

- Quantify catalog composition and trend behavior.
- Identify high-leverage countries, ratings, and genres.
- Generate actionable recommendations from visual diagnostics.

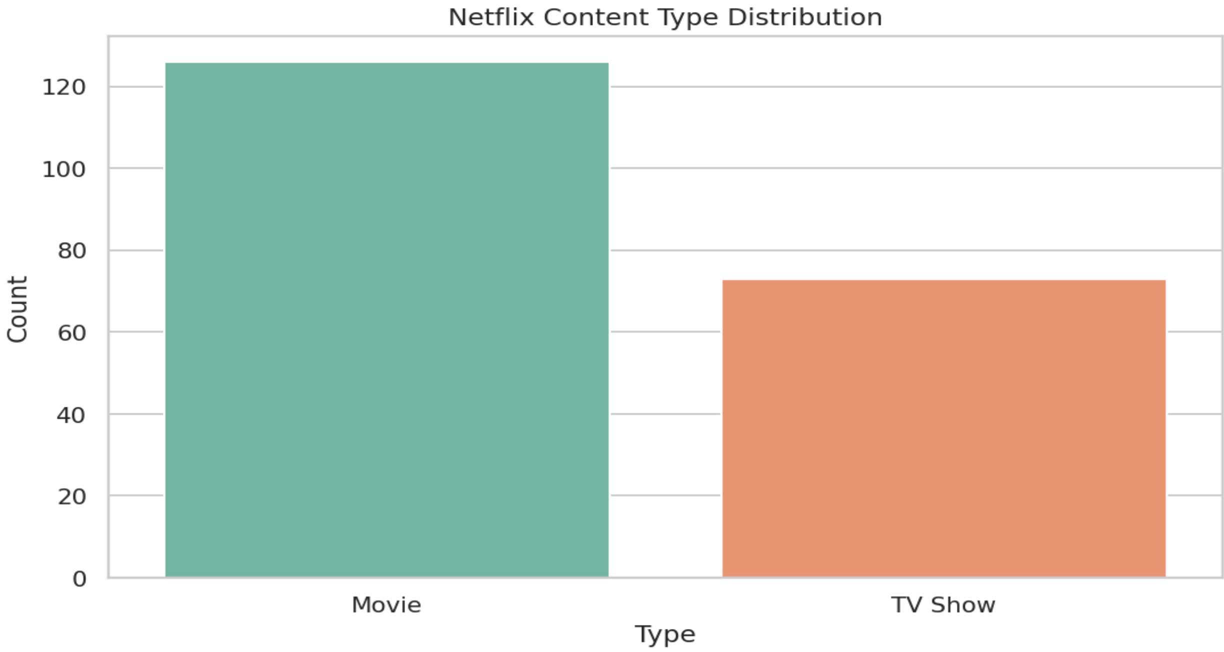
Dataset Overview

Attribute	Value
Rows	199
Columns	16
Movie Share	63.32%
TV Show Share	36.68%
Missing Value %	2.61%

Data Quality and Preparation

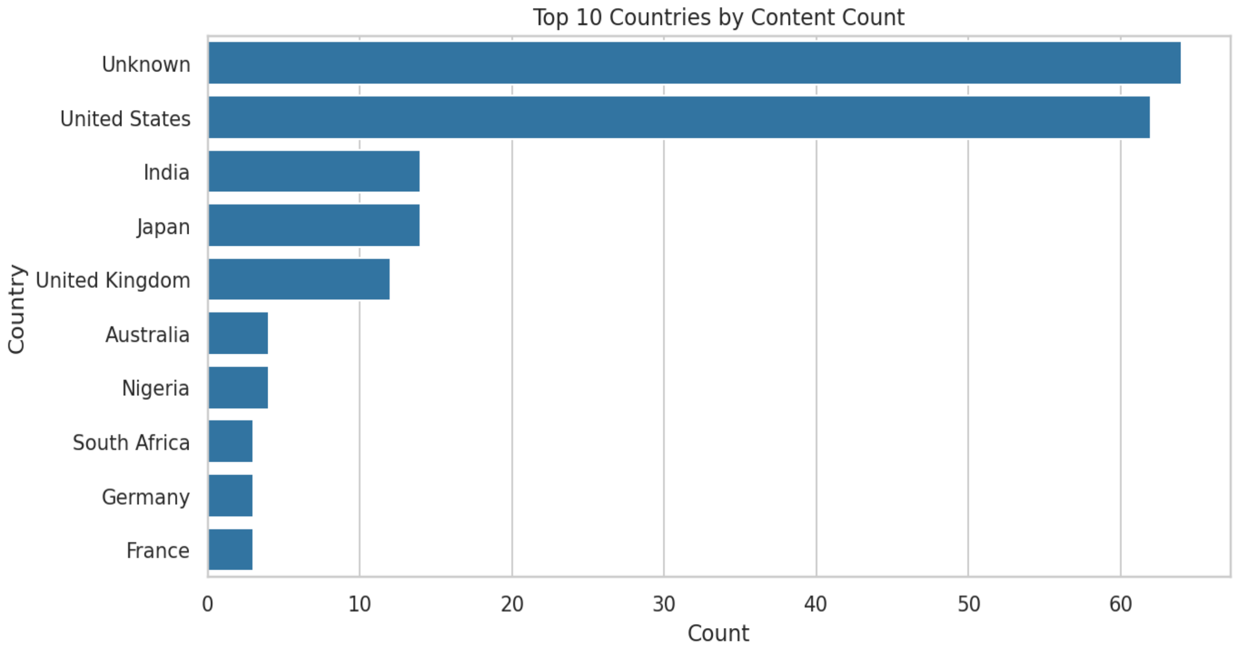
- Date, release-year, and duration fields were normalized for analysis readiness.
- Primary-country and primary-genre derived for interpretable grouping.
- Missing metadata retained but surfaced as quality risk in recommendations.

Movies vs TV Shows Distribution



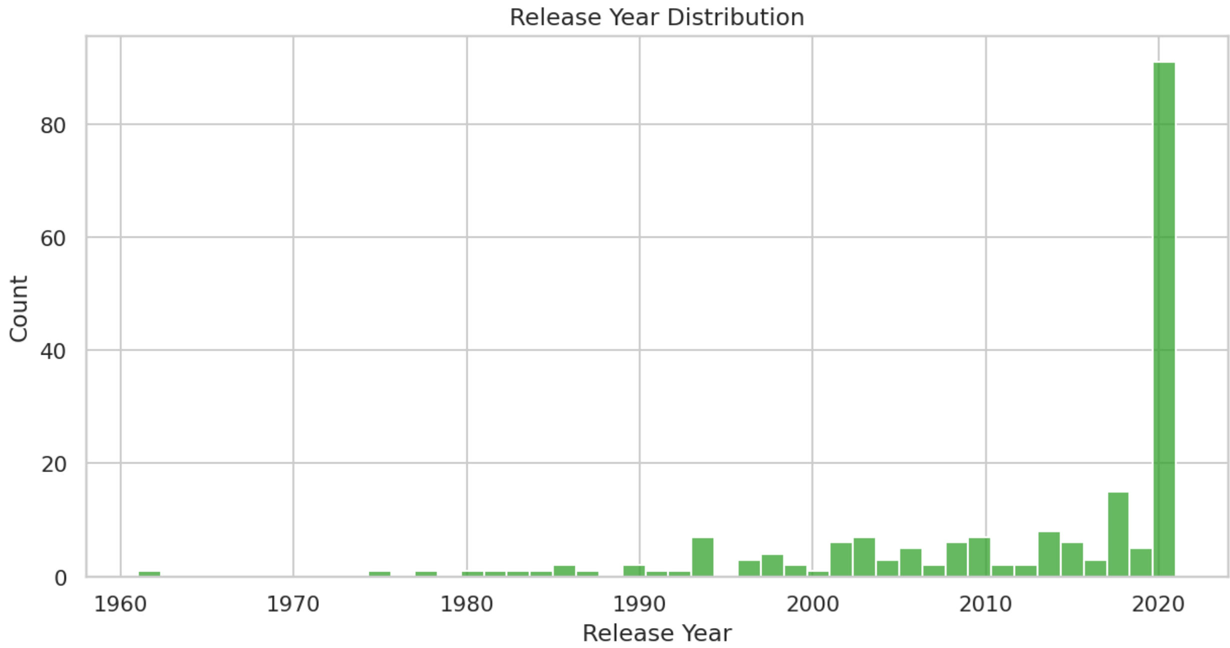
• Content-type composition indicates the platform's portfolio balance.

Top Content-Producing Countries



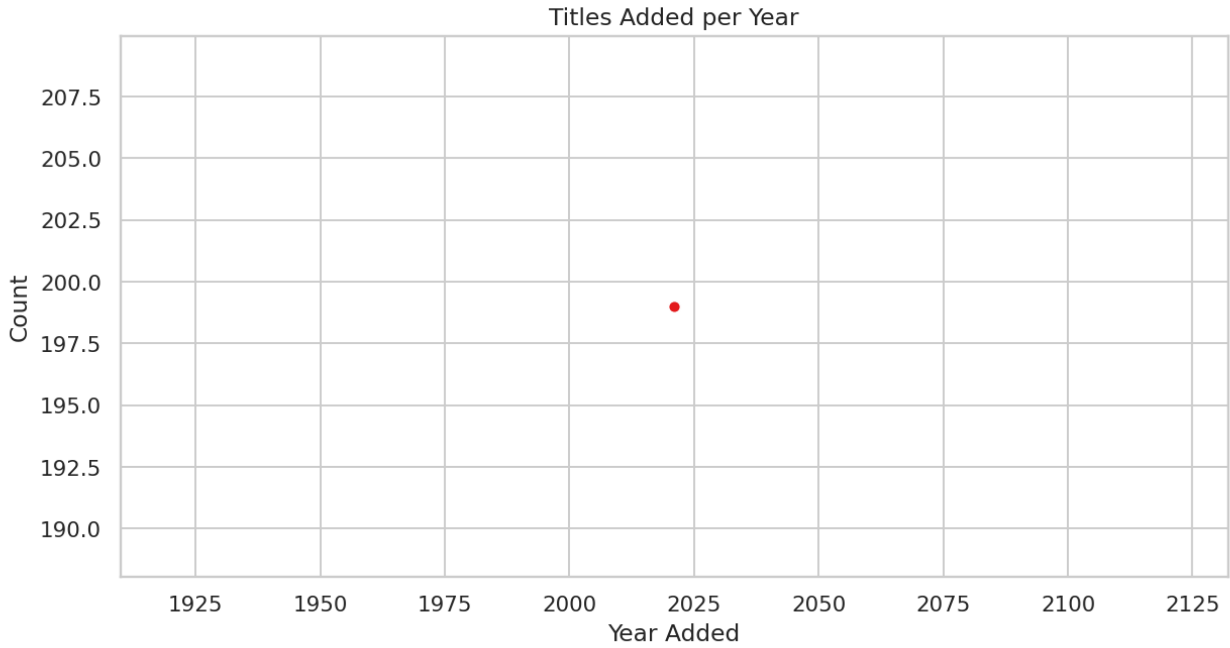
• Country concentration informs regional diversification strategy.

Release Year Distribution



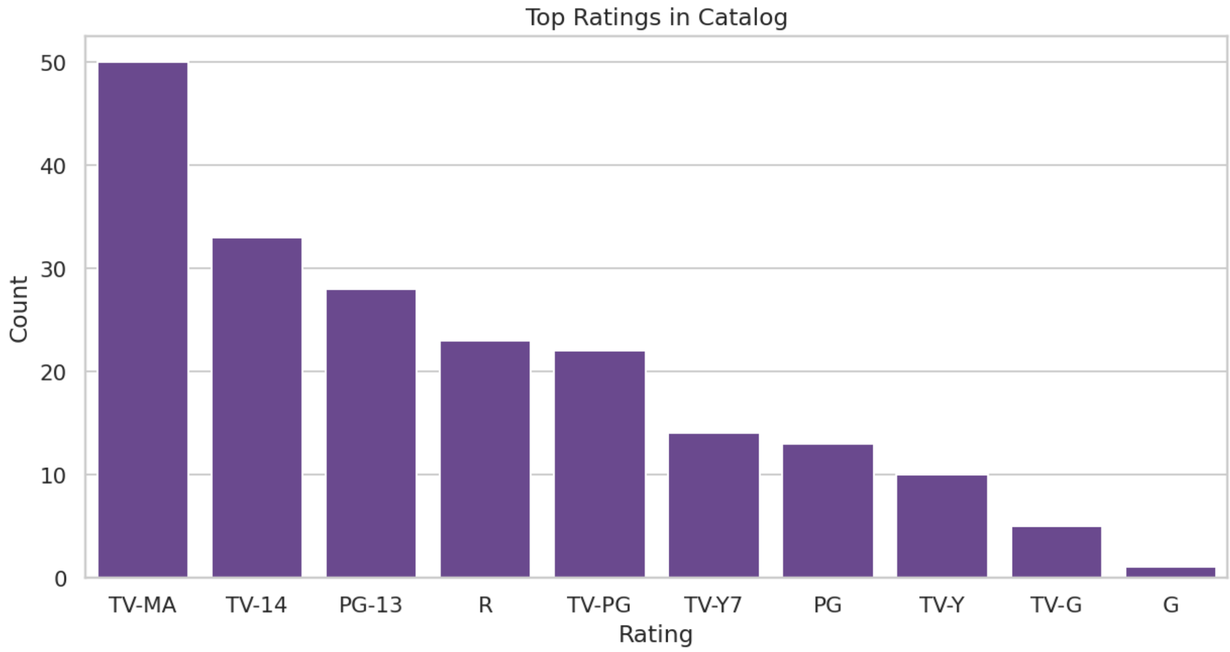
• Catalog skew toward recent years highlights recency-driven curation.

Content Added Over Time



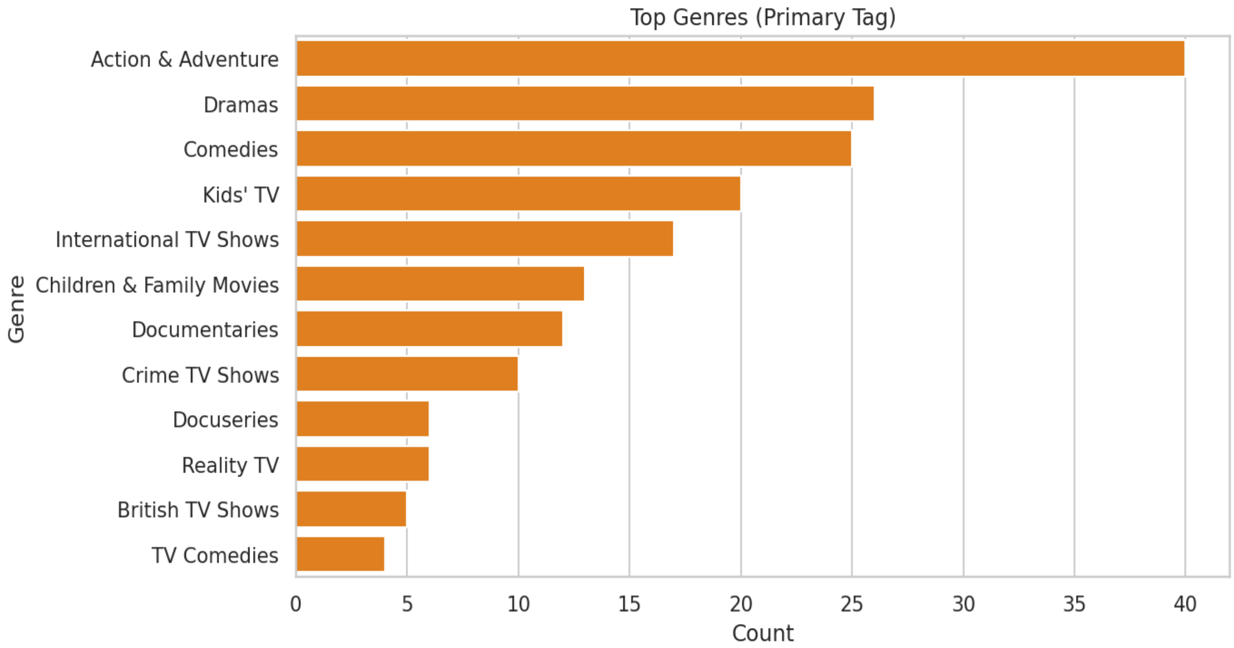
• Additions over time show platform growth phases and cadence shifts.

Rating Distribution



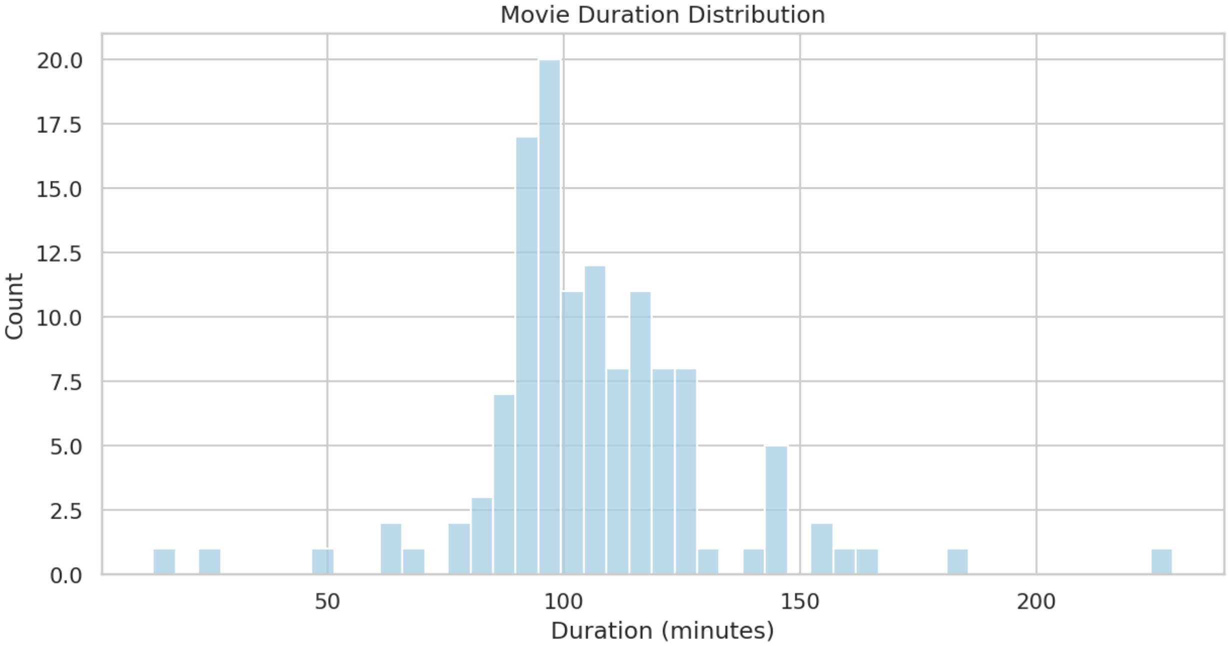
• Rating concentration indicates dominant audience maturity segments.

Top Primary Genres



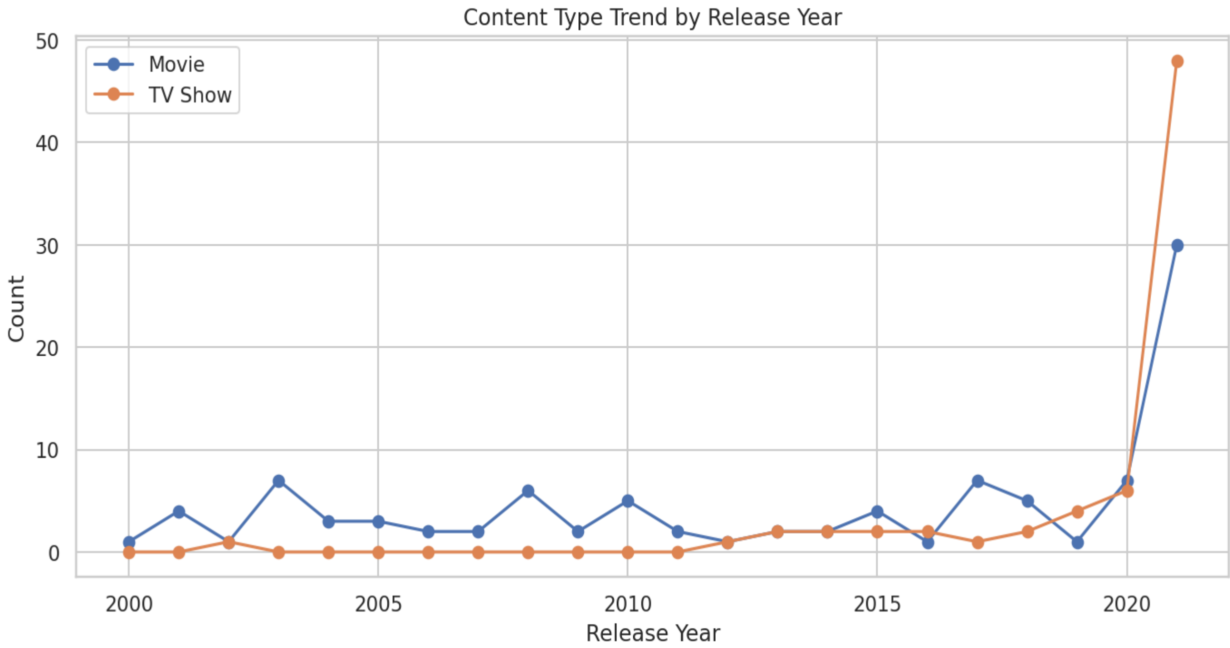
• Genre dominance guides content investment and recommendation weighting.

Movie Duration Distribution



• Duration profile supports scheduling and user attention modeling.

Content-Type Trend by Release Year



• Type trend evolution highlights shifts in content strategy emphasis.

Statistical and Reporting Summary

Metric	Result
Top Country by Titles	Unknown
Median Release Year	2,018
Most Common Rating	TV-MA
Most Common Genre	Action & Adventure
Average Movie Duration	106.05

Key Insights

- Catalog composition and trend analysis reveal strategic concentration zones.
- Country and genre dominance suggest opportunities for diversification.
- Metadata quality directly affects recommendation-system utility.

Business Recommendations

- Balance movie/series investments using demand and retention objectives.
- Expand underrepresented regional content clusters.
- Prioritize metadata completeness for better discovery and personalization.

Implementation Roadmap

- 0-30 days: establish catalog health dashboard with type/region/rating metrics.
- 30-60 days: launch targeted content-gap acquisition sprint.
- 60-90 days: integrate metadata quality KPIs into content pipeline governance.

Risks and Limitations

- Country and genre fields can be multi-valued and noisy.
- Catalog snapshots may not capture engagement-level outcomes.
- Business impact depends on alignment with subscriber behavior metrics.

Reproducibility and References

- Netflix titles dataset used in business-case workflow
- Notebook lineage: Netflix case_stiudy .ipynb
- Method stack: pandas, seaborn, exploratory visualization

Chapter Closure

- EDA establishes a defensible baseline for content portfolio decisions.
- Visualization-led insights convert complex catalog structure into strategic direction.